

TOPPR PERSONAL STUDY VAULT

Plant Pathology

Section A Overview & Study Guide

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Section Study Notes

Study Notes: Plant Pathology (3201) - Section A (Sabiha Mam)

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Lecture: 1.0 Introduction and History of Plant Pathology

Course Slides: 1.0 Introduction and History of Plant Pathology

Slide 1

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Slide 2

What is plant pathology ?

Plant Pathology (Phytopathology): Greek Word

Three words: Phytos + Pathos + logos

Plant + suffering/ailments + study/knowledge

Definition: Plant Pathology is that branch of agricultural, botanical or biological science.

Slide 3

What is plant pathology ?

Plant pathology deals-

Study of the causes, etiology, resulting losses and management/ control of plant diseases.

Study of the organisms and environmental factors that cause diseases in plants, mechanisms by which they cause disease, and methods of managing diseases and reducing damage.

Slide 4

Why do we study Plant Pathology?

To study the biotic, abiotic and environmental causes of plant diseases

To study the interaction(s) between the plant and the pathogen

To study the mechanism(s) of disease development.

To develop management strategy of the diseases

Slide 5

What is the importance of Plant Pathology in Agriculture ?

Plant pathology can and should contribute in each of these respects—

by assessing the immediate and potential dangers to crops from diseases

by forecasting their incidence and severity

by deploying the best methods of control in the short and longer term

by evaluating the risks particularly by integrated use of the management practices.

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History of Plant Pathology

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History of Plant Pathology

Four Phases

Ancient period

Dark period

Renaissance period

Modern period

Slide 8

Ancient Period

Theophrastus (300-286 B.C.)

Study and write about the diseases (rust, mildews, blight) of trees, cereals and legumes in his book enquiry into plants.

His approach was observational and speculative rather than experimental.

Theory of spontaneous generation

2 Books : Historia Plantarum

De Causis Plantarum

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Father of botany

Slide 9

Albertus Magnus (1200 A.D.)

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Recognized Mistletoe as the first plant pathogen and the first pathogen for which a cultural control (by pruning affected branches) was recommended.

Slide 10

DARK PERIOD

300AD and 1300AD

Also called pre-renaissance period in history

No increase in the knowledge of plant pathology

An Arabian Ibnal Awan described symptoms & control measures of many trees & vine diseases.

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PRE-MODERN/ RENAISSANCE PERIOD/AUTOGENIC ERA (17th –mid 19th)

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1675 A. D.-Anton van Leeuwenhoek (Holland)

Invented compound microscope in 1675

Examined anatomy of plants and body of filamentous fungus, algae, protozoa, sperm cells, blood cells and bacteria.

In 1683 he described bacteria seen with this microscope

Father of Microbiology

Slide 12

1729 A. D.-Pier Antonio Micheli (Italian)

Observed fungal spores for the first time and conducted studies related to many spores germination in 1729

He published a book “Nova Plantarum Genera” in which he gave descriptions about 1900 species in Latin out of which 900 were fungi.

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[Image not found: /uploads/1.0 Introduction and History of Plant Pathology_s12_img2.jpg]

Father of Mycology

Slide 13

1755 A. D.-Tillet (France)

Published a paper on bunt or stinking smut of wheat

He reported the chemical treatment of seeds

Slide 14

MODERN PERIOD/ GOLDEN ERA/PATHOGENIC ERA (1800 ---))

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IRIS FAMINE

In 1845, the potato crop in Ireland was completely wiped out by late blight disease

Caused great famine in 1846.

This resulted in the death of hundreds of thousands of people

Immigration of more than one and a half million Irish to United States.

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Irish famine Monument

Slide 15

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Anton De Bary (1853 A. D).Germany)

Established that fungi are the causes, not result of plant diseases

Renamed the casual organism of late blight as *Phytophthora infestans*.

He discovered heteroecious nature of rust fungi

Detailed life cycles of downy mildew

He wrote a book named "Morphology and Physiology of fungi, lichens and Myxomycetes" (1866).

Father of Modern Mycology

Slide 16

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Louis Pasteur

Established the germ theory

In 1864, Pasteur invented pasteurization

in 1880, made the first vaccine against the chicken cholera.

Germ theory

Microorganisms were produced from preexisting microorganisms

Most infectious diseases were caused by germs.

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Robert Koch (1879 A. D.)

Established Koch's postulates(rules of disease diagnosis)

In 1876 identified the anthrax bacillus, *Bacillus anthracis*, as the first bacterium to cause disease in animals and humans.

Father of Bacteriology

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[Image not found: /uploads/1.0 Introduction and History of Plant Pathology_s18_img2.jpg]

Bacillus anthracis

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1889 A. D. Beijerinck (Dutch)

proved that the virus is not a living microorganism.

N.A. Cobb

Contributed a lot to nematode morphology, taxonomy and methodology

Founder of Virology

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Founder of Nematology

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Paul Neergaard

He helped to standardize methods for the detection of seed borne fungi.

"Seed Pathology" (1977) which served as a reference and teaching standard for the science of seed pathology throughout the world.

Father of seed pathology

Slide 22

Plant Pathology

Bangladesh Perspective

Slide 23

Bengal Famine 1943 (more 3-4 million people died)

One of the causes of famine was Brown spot disease of rice

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[Image not found: /uploads/1.0 Introduction and History of Plant Pathology_s23_img2.png]

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Bangladesh perspective

Department of plant pathology

Seed Pathology Centre

Plant Disease diagnostic Clinic

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<div id="1-1-pathogenesis-infection-process"></div>

Lecture: 1.1 Pathogenesis & Infection Process

Course Slides: 1.1 Pathogenesis

Slide 1

Chapter 1

Pathogenesis

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Slide 2

Intended Learning Outcomes

Differentiate parasitism and pathogenicity

State sequential events in disease development

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Slide 3

Q

Parasite: An organism that lives on and derives food from another organism.

Parasitism: The relationship between a parasite and its host is parasitism.

Pathogen: An agent that can incite a disease.

Pathogenicity: It is the ability of the pathogen to cause disease.

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Slide 4

Parasitism and Pathogenicity

Most successful pathogens

fungi,

bacteria,

mollicutes,

parasitic higher plants,

nematodes,

protozoa,

viruses and

viroids

Slide 5

Q

Where the viruses, viroids, mycoplasmas, and fastidious bacteria overseason?

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Slide 6

Parasitism and Pathogenicity

Are all pathogens parasites?

A parasite is not necessarily a pathogen and a pathogen is not necessarily a parasite

But why?

Slide 7

Parasitism and Pathogenicity

Symbiosis such as mycorrhizae or root nodule bacteria.

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Slide 8

Symbiotic Relationship is a type of parasitism, explain.

Q

Symbiotic relationship refers to any close and long-term interaction between two organisms, including mutualism, commensalism, and parasitism.

Parasitism is a specific type of symbiosis where one organism, the parasite, benefits by exploiting the host for nutrients, water, or shelter, causing harm to the host. Pathogen like fungi (*Puccinia graminis*) is a plant parasite. These pathogens attach to, penetrate, infect, and invade host tissues, leading to symptoms such as necrosis, wilting, and reduced yield.

While all parasitism is symbiotic, not all symbiosis is parasitic, as mutualism and commensalism do not harm the host.

Slide 9

Parasitism and Pathogenicity

Biotrophs/Obligate Parasites

Non-obligate (2)

Facultative parasite/Necrotrophs.

Facultative saprophytes/Hemibiotrophs /semibiotrophs

Slide 10

Q

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Obligate Parasites

Slide 11

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Q

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Characteristics of Necrotrophs

Biochemical and morphological

features:

Host cell rapidly killed

Toxins and cytolytic enzyme produced

No special parasitic structures formed

Host penetration via wounds or natural openings

Slide 13

Characteristics of Necrotrophs

Ecological features

Wide host range

Able to grow saprophytically away from the host

Attack juvenile, debilitated or senescing tissues

Slide 14

Characteristics of Biotrophs

Biochemical and morphological features:

Host cells not rapidly killed

Few or no toxins or cytolytic enzymes produced

Special parasitic structures eg, haustoria are formed

Host penetration direct or via natural openings

Slide 15

Characteristics of Biotrophs

Ecological features

Narrow host range

Unable to grow away from the host

Attack healthy hosts at all stages of development

Slide 16

Pathogenesis/Disease cycle

The chain of events involved in disease development is called a disease cycle or pathogenesis

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Q

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[Image not found: /uploads/1.1 Pathogenesis_s17_img1.png]

Q

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Q

Stages in the development of disease

Inoculation

Pre-penetration Phenomena

Penetration

Infection

Invasion

Growth and Reproduction of the Pathogen

Dissemination of the Pathogen

Over-wintering/over-summering of the Pathogen

Slide 19

Q

Pathogenesis: The chain of events involved in disease development, including the stages of development of the pathogen and the effect of disease on the host is called disease cycle or pathogenesis.

What Happens If Colonization Doesn't Occur In Disease Cycle ??

Disruption of disease cycle

No establishment of pathogen

Colonization: In Pathogenesis, It refers to the successful establishment and multiplication of the pathogen within the host tissue after the infection has occurred.

Limited pathogen spread

No infection or disease development

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Pathogen survival is jeopardized

Without colonization and the subsequent growth and reproduction of the pathogen, the disease cycle is effectively interrupted resulting the host remains unaffected, the pathogen fails to thrive, and transmission to other hosts does not occur, ultimately leading to the pathogen's diminished survival and spread.

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Q

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Q

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Slide 22

Q

Pathogen:

An entity that can incite disease.

Inoculum:

The pathogen or its parts that can cause infection; that portion of individual pathogens that are brought into contact with the host.

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Inoculation

Inoculation

Inoculum

Propagule

Slide 24

Which are Inocula?

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Fungus

Bacteria, mollicutes, protozoa, viruses and viroids.

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[Image not found: /uploads/1.1 Pathogenesis_s24_img4.jpg]

Slide 25

Which are Inocula?

Nematodes

Phanerogamic parasites

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[Image not found: /uploads/1.1 Pathogenesis_s25_img2.jpg]

Slide 26

Q

Types of Inoculum

Primary inoculum and primary infection (Carries itself to the host body by itself)

Secondary inoculum and secondary infection (Produced in the affected plant)

Slide 27

Sources of inoculum

Inside sources

Outside sources

Slide 28

Arrival or landing of inoculum on host

Most is carried by wind, water, and insects.

Only a small number actually lands on susceptible plants.

May be carried by vectors.

Slide 29

Pre-penetration Phenomena

Slide 30

Q

i) Attachment of pathogen to host

The propagules of pathogen secrete mucilaginous substances which helps to adhere to the plant.

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ii) Spore germination and perception of the host surface

iii) Appressorium Formation and Maturation

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iv) Recognition between Host and Pathogen

The components released from both of host and pathogen are responsible for the recognition.

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Attachment of Pathogens to Host

Viruses, mollicutes, fastidious bacteria, and protozoa.

Almost all fungi, bacteria and parasitic plants.

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Attachment of Pathogens to Host

The propagules have a mucilaginous layer.

The germ tubes also produce these substances.

The areas of contact appear to degrade due to enzymes.

Slide 33

Germination of spores and seeds

Requirements for germination

Favorable temperature

Favorable moisture in the form of rain, dew, or a film of water

Slide 34

Pre-penetration Phenomena

Germination of spores and seeds

Spore germination is often favored by

Exudates

Certain pathogens are only stimulated by exudates of plants susceptible to that particular pathogen.

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Pre-penetration Phenomena

Germination of spores and seeds

Spore germination may be inhibited by

materials released by the plant

by saprophytic microflora

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Pre-penetration Phenomena

Pre-penetration Phenomena

Germination of spores and seeds

Antagonistic microorganisms

Fungistasis

Suppressive soils

Slide 37

Prepenetration Phenomena

Pre-penetration Phenomena

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[Image not found: /uploads/1.1 Pathogenesis_s37_img2.png]

A rust uredospore that has germinated and produced a dome-like appressorium.

Uredospores of a rust fungus next to open stomata.

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Pre-penetration Phenomena

Pre-penetration Phenomena

After spores germinate

the resulting germ tubes must grow or

the secondary zoospores must move toward a penetration site

The number, length and rate of growth of germ tubes or the number and mobility of motile spores affected by temperature and moisture etc.

Slide 39

Prepenetration Phenomena

Pre-penetration Phenomena

Chemical stimuli associated with wounds, stomata and lenticels also stimulate growth

Seeds germinate by producing a radicle

Slide 40

Hatching of Nematode Eggs

Favorable temperature and moisture

The egg contains the first juvenile stage before or soon after the egg is laid.

This juvenile immediately undergoes a molt and gives rise to the second juvenile stage, which may remain dormant in the egg for various periods of time.

Slide 41

Hatching of Nematode Eggs

Thus, when the egg finally hatches, it is the second-stage juvenile that emerges, and it either finds and penetrates a host plant or undergoes additional molts that produce further juvenile stages and adults.

Slide 42

Hatching of Nematode Eggs

They are attracted to the root by CO₂ and some amino acids associated with root growth

Slide 43

Q

Penetration

Pathogens penetrate plant surfaces through natural openings or wounds or directly

Some fungi penetrate only in one way, others in several ways

Bacteria enter mostly through wounds sometimes through natural openings but never directly

Viruses, viroids, mollicutes fastidious bacteria and protozoa enter through wounds by vectors

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Q

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Q

[Image not found: /uploads/1.1 Pathogenesis_s45_img1.jpg]

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Q

Direct penetration through intact plant surfaces/ Use of physical weapons

Fungi, nematodes and parasitic plants.

Fungi use fine hyphae.

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Q

Direct penetration through intact plant surfaces

The penetration is through mechanical force and softening of the cell walls by an enzyme.

Most fungi form an appressorium at the end of the germ tube.

Then a penetration peg emerges from the flat surface of the appressorium and pierces the cuticle and cell wall.

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Q

Direct penetration through intact plant surfaces

The peg grows into fine hyphae then reaches a normal diameter once it is inside the cell

Parasitic plants also penetrate the same way

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Q

Direct penetration through intact plant surfaces

Nematodes penetrate its stylet

Inserts its stylet or the entire nematode enters

[Image not found: /uploads/1.1 Pathogenesis_s49_img1.png]

Slide 50

Penetration through wounds

Bacteria, most fungi, some viruses, all viroids

Viruses mollicutes, fastidious vascular bacteria, and protozoa enter through wounds made by vectors

Some pathogens can be only carried by specific vectors

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Q

Penetration through natural openings

Many fungi and bacteria enter through:

Stomata

Hydathodes

Nectarhodes

Lenticels

Slide 52

Q

[Image not found: /uploads/1.1 Pathogenesis_s52_img1.jpg]

Slide 53

Penetration through stomata

Underside of the leaf

Bacteria swim

Fungi can germinate on a wet surface and enter a stomata

Slide 54

Hydathodes

Open pores on the margins and tips of leaves

Connected to veins

Secrete droplets of liquids containing nutrients some bacteria but few fungi enter here

Some enter through nectarhodes which are similar to hydathodes but on blossoms

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Lenticels

Openings on fruits, stems, and tubers, filled with loosely connected cells to allow passage of air

Relatively few fungi and bacterial enter this way

A less efficient, secondary pathway

Slide 56

Path of Entry

Fungi = direct, wounds, natural openings

Bacteria = wounds, natural openings

Nematodes = direct, natural openings

Parasitic higher plants = direct

Virus, viroids, mollicutes, protozoa and fastidious bacteria =wounds

Slide 57

Q

Infection

Establishment of contact by the pathogens with the host plant is called infection.

During infection, the pathogen grows or multiplies, colonizes the host plant

Infection results in the appearance of symptoms

Some infections remain latent and show up when conditions are more favorable

Slide 58

Infection

Symptoms may show up in 2-4 days or as long as 2-3 years

Incubation period

In most plants the incubation period is from a few days to a few weeks

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Infection

During infection some pathogens:

Obtain nutrients without killing the cell

Kill cells and use contents

Kill cells and disorganize surrounding tissue

Release enzymes, toxins, growth regulators

Slide 60

Q

Successful infection may or may not always result in appearance of symptoms, explain with examples:

A successful infection occurs when a pathogen invades a host plant, establishes itself, and begins to multiply. However, this does not always lead to visible symptoms of disease.

Scenarios Where Successful Infection May Not Cause Symptoms

Latent Infections:

Pathogens infect the host and remain dormant or inactive for some time before symptoms appear.

Example: Apple's latent virus.

Slide 61

Q

2. Tolerant Host Plants:

Some plants are infected but can resist or tolerate the pathogen's presence without showing symptoms.

Example: Some wheat varieties can host Barley yellow dwarf virus.

3. Environmental Conditions:

The environment may not be conducive to disease development even if infection occurs.

Example: *Phytophthora infestans* (late blight pathogen) may infect potatoes, but symptoms won't develop in dry conditions.

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Q

Invasion

The spread of the pathogen into the host is called invasion.

Apple scab grows between the cuticle and the epidermis

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Invasion

[Image not found: /uploads/1.1 Pathogenesis_s63_img1.png]

Powdery mildew grow on the surface and send haustoria into epidermal cells

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[Image not found: /uploads/1.1 Pathogenesis_s64_img1.png]

[Image not found: /uploads/1.1 Pathogenesis_s64_img2.png]

intercellularly

[Image not found: /uploads/1.1 Pathogenesis_s64_img3.png]

Hyphae of the smut fungus *Ustilago* in an infected leaf (intracellularly)

vascular wilts invade the xylem vessels

(Both fungus and bacteria)

Slide 65

Invasion

Fungi invade intercellularly and intracellularly and some grow into and throughout the plant

Bacteria at first invade intercellularly and then intracellularly.

Bacteria causing vascular wilts, invade the xylem vessels

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[Image not found: /uploads/1.1 Pathogenesis_s66_img1.png]

[Image not found: /uploads/1.1 Pathogenesis_s66_img2.png]

Alfalfa shoot invaded by plant parasitic nematodes (*Ditylenchus dipsaci*)

Slide 67

Invasion

Most nematodes are intercellular some intracellular and some don't invade at all but feed by piercing epidermal cells with their stylets.

Viruses, viroids, mollicutes, fastidious bacteria, and protozoa invade by moving from cell to cell

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Types of cells and tissues invaded

Fungi, viruses, and viroids invade all types of cells

Mollicutes and protozoa invade phloem sieve tubes and some parenchyma

Most fastidious bacteria invade xylem vessels and a few invade phloem sieve tubes

Slide 69

Local and Systemic invasion

Fungi, bacteria, nematodes, viruses, and parasitic plants : local

Fastidious bacteria, mollicutes, and protozoa: systemic (internal)

Some fungi, bacteria, and viruses are also systemic

Slide 70

Difference between infection and Invasion

[Image not found: /uploads/1.1 Pathogenesis_s70_img1.png]

Slide 71

Growth and Reproduction of the Pathogen

Fungi= spores, sexual or asexual

Parasitic plants= seeds on branches

Bacteria, mollicutes, protozoa = fission

Viruses and viroids= replicate

Nematodes = eggs

Slide 72

Growth and Reproduction of the Pathogen

Location of reproduction

Most fungi reproduce inside the plant but most release spores outside

Powdery mildew on the outside

Viruses, viroids, mollicutes, protozoa, and fastidious bacteria only reproduce inside

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[Image not found: /uploads/1.1 Pathogenesis_s73_img2.png]

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Growth and Reproduction of the Pathogen

Rate of reproduction

Fungi: millions of spores in a season

Bacteria divide every 20-30 minutes

Viruses can produce 100,000 to 10 million particles in a single cell

Nematodes lay 300-500 eggs about half female who do the same and so on

2 to 8 million, nematodes are reproduced in one season during one season

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[Image not found: /uploads/1.1 Pathogenesis_s75_img2.png]

Slide 76

Dissemination

The Spread of the pathogen from the sources of the pathogen to the healthy plant is called dissemination.

[Image not found: /uploads/1.1 Pathogenesis_s76_img1.png]

Slide 77

Dissemination by air

Fungal spores and seeds

The spores land or are washed out by rain

Spores can be carried from several to several hundred kilometers (high altitude)

This can cause an epidemic over several years. To stop this, wind break may be used.

Bacteria and nematodes may be carried by wind, water, etc.

Slide 78

Wind as a means of dispersal

Short distance dissemination

Sporangia of downy mildew fungi, conidia of powdery mildew fungi and basidiospores of rust fungi

Long Distance

Uredospores of rust fungi, Chlamydospores of smut fungi and conidia of Alternaria, Helminthosporium and Pyricularia,

Slide 79

Dissemination by air

Bacterial fire blight exudes

Nematodes and spores with ground debris

Windy rain

Touching plants

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Slide 80

Dissemination by water

Bacteria, nematodes, spores, mycelium parts in the soil

All Bacteria and many spores are exuded in a sticky liquid

Rain drops or drops from overhead irrigation

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Dissemination by Insects, Mites, and Other Vectors

Aphids, leafhoppers, whiteflies.

Leafhoppers: mollicutes, fastidious bacteria, and protozoa.

Dutch Elm disease and bacterial wilt of cucurbits.

Soft rot, anthracnose, and ergot are carried by insects externally.

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Dissemination by Insects, Mites, and other Vectors

Mites and nematodes can also carry virus pathogens internally and bacteria and fungus externally

Animals that walk among plants also carry

Parasitic plants can carry as they bridge

[Image not found: /uploads/1.1 Pathogenesis_s82_img1.jpg]

Slide 83

Dissemination by Seed, Transplants, Budwood, and Nursery Stock

During propagation

The grower can infect his own stock and sell it out to garden centers, home owners etc.

Ex: Crown gall

Slide 84

Dissemination by Humans

Handling diseased stock

Tobacco mosaic is transmitted through cigarettes, shoes, hands

Traveling long distances

Tools carry pathogens from plant to plant

Fireblight is a good example

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Dissemination by Humans

Dutch elm disease fungus

White pine blister rust fungus

Citrus canker bacterium

Powdery and downy mildew fungus of grape (Europe)

<div id="1-2-toxin-and-host-defense"></div>

Lecture: 1.2 Toxin and Host Defense

Course Slides: 1.2 Toxin

Slide 1

The Effect Of Toxins

in Plant Tissue

[Image not found: /uploads/1.2 Toxin_s1_img1.png]

[Image not found: /uploads/1.2 Toxin_s1_img2.png]

[Image not found: /uploads/1.2 Toxin_s1_img3.jpg]

Slide 2

Toxin

Toxins are injurious/toxic substances secreted by micro-organism host complex that acts on living host protoplast to affect disease symptoms.

Death of cells : De Bary in 1886

Oxalic acid secreted by Sclerotinia spp. Killing of cells.

Any substance produced by a pathogen, that is injurious to the host and plays a crucial role in pathogenesis is called toxin. Toxins involved in plant parasite interactions are also called aggression.

Chemical Weapon of Pathogen

These are simple molecules with low molecular weight, they are extremely mobile and easily reach the sub cellular level of the host.

Q

Slide 3

[Image not found: /uploads/1.2 Toxin_s3_img1.jpg]

Q

Slide 4

Classification

Exo-Toxins

Endo-Toxins

Toxins

Role of Pathogenesis

Specificity to Host

Vivo toxins

(Diamond & Waggoner, 1953)

Pathotoxins

(Wheeler and Luke, 1963)

Phytotoxins

Role is suspected in disease

Alternaria; Alternaria;

cochliobolin.

Causal role in the disease

Tab-toxin;

HMT toxin

Partial role in disease

Fusaric acid

Pyricularin

Tab toxin;

phseolotoxin,

Tentoxin

Cercosporin

Victorin

T-toxin,

HC-toxin,

AAL-toxin

Slide 5

Q

Types of toxin in Pathogenesis

Vivotoxin

A vivotoxin is defined as a substance produced in the infected host by the pathogen and/or its host, which functions in the production of disease but is not itself the initial inciting agent of disease.

Characteristics of vivotoxin:

Reproducible separation of the toxin from the sick plant

Purification or chemical characterization

Induction of at least a portion of the disease symptoms by placing the toxin in a healthy plant.

Typical examples are fusaric acid and pyricularin.

Slide 6

Q

Types of toxin in Pathogenesis

Pathotoxin:

A chemical of biological origin.

Host specific toxins that produces almost all the disease symptoms in absence of causal organism. Eg. T toxin, Victorin, Myrticin, HC toxin etc.

Phytoalexins in plants:

Low molecular weight antimicrobial compounds that are produced by plants as a response to biotic and abiotic stresses.

Take part in an intricate defense system which enables plants to control invading microorganisms.

Slide 7

Victorine is a host specific toxin

Q

[Image not found: /uploads/1.2 Toxin_s7_img1.png]

Slide 8

Q

Phytotoxins:

Phytotoxins are poisonous substances

Toxins synthesized by plants or plant pathogens through naturally occurring biochemical reactions.

Plants use them as defensive agents against predators (pests, insects, stress, etc.) and diseases

Slide 9

Q

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Slide 10

Q

Effect of Toxins on Plant Tissue

mechanisms of phyto-toxicity

Slide 11

Effect of Toxins on Plant Tissue

Changes in Cell Wall Permeability

Slide 12

Changes in cell wall permeability:

Toxins kill plant cells by altering the permeability of the plasma membrane thus permitting

Loss of water and

The electrolyte leakage/ loss of electrolytes and

unrestricted entry of substances including the toxin.

Na⁺, Ca²⁺, K⁺, Cl⁻, PO₄³⁻, and Mg²⁺

The electrolyte leakage is ubiquitous among different species, tissues, and cell types, and can be triggered by all major stress factors, including pathogen attack (Atkinson et al., 1985, 1990, 1996; Ebel and Mithofer, 1998; Blatt et al., 1999; Maffei et al., 2007)

Slide 13

[Image not found: /uploads/1.2 Toxin_s13_img1.jpg]

Toxins probably either destroy them or interfere with the mechanisms involved in their synthesis.

Affect cellular transport system of H⁺/K⁺ exchange in cell membrane.

DEATH

For example:

Lycomarasmin, fusaric acid, picolinic acid, victorin und other toxins.

Victorin can cause increased permeability at very low concentrations.

Slide 14

Effect of Toxins on Plant Tissue

2. Disruption Of Normal Metabolic Processes

Slide 15

Disruption of normal metabolic processes

Role of Potassium Ion (K⁺)

catalyzes chemical reactions by regulating > 60 enzymes

acts as a catalyst by activating enzymes involved in regulating the rate of photosynthesis and the production of ATP.

regulate electrical charge, which is important to maximize ATP production during photosynthesis.

Regulates Stomata Opening & Closing Regulates Stomata Opening & Closing

K

+

Slide 16

Disruption of normal metabolic processes

Slide 17

Effect of Toxins on Plant Tissue

3. Malfunctioning of Enzyme System

Slide 18

Tabtoxin, Phaseolotoxin and Tentoxin

Inhibit normal host enzymes, thereby leading to increases in toxic substrates or to depletion of needed compounds

[Image not found: /uploads/1.2 Toxin_s18_img1.jpg]

[Image not found: /uploads/1.2 Toxin_s18_img2.png]

[Image not found: /uploads/1.2 Toxin_s18_img3.png]

Tabtoxin Phaseolotoxin Tentoxin

Slide 19

Tabtoxin:

Dipeptide compound of amino acid threonine and tabotoxinine

Tabotoxinine

Active form

Toxic to cell

It inactivate the enzyme

glutamine synthetase

[Image not found: /uploads/1.2 Toxin_s19_img1.png]

Slide 20

Q

Some non-toxic chemicals can become toxic due to their conversion into active toxins inside the plant. This process typically involves enzymatic reactions that occur within the host plant. Tabtoxin is a classic example of this mechanism.

Tabtoxin and Its Toxicity Mechanism

Produced by Pathogen:

The bacterium *Pseudomonas syringae* pv. *tabaci* produces tabtoxin, a non-toxic precursor molecule.

Tabtoxin itself is not harmful to plants in its original form.

Structure of Tabtoxin:

Tabtoxin consists of tabtoxinine- β -lactam linked to a dipeptide structure.

This structure allows it to remain non-toxic initially.

Slide 21

Q

Activation in the Host:

Once tabtoxin enters the plant cells, plant enzymes break it down, releasing the active component called tabtoxinine- β -lactam.

This breakdown process occurs naturally within plant metabolism.

Toxic Effect:

Tabtoxinine- β -lactam inhibits the enzyme glutamine synthetase.

This inhibition prevents the conversion of ammonia (NH_3) to glutamine, causing a toxic build-up of ammonia in plant tissues.

Slide 22

After Inactivation of Glutamine Synthetase:

Depleted glutamine levels

Accumulation of toxic concentrations of ammonia

Uncouples photosynthesis and photorespiration

Destroy the thylakoid membrane of the chloroplast

Chlorosis and necrosis

Slide 23

Phaseolotoxin:

The toxin that affects cells by binding to the active site and inactivating the enzyme ornithine carbamoyl transferase.

Accumulate ornithine

Depleted levels of citrulline and arginine

Slide 24

Tentoxin:

Inhibits the activity of polyphenol oxidases, enzymes involved in several resistance mechanisms of plant.

Other examples:

Pyricularin inhibits polyphenol oxidase system and

Victorin acts by uncoupling of oxidative phosphorylation.

Fusaric acid and picolinic acid can form chelating complexes with heavy metal ions and could inhibit enzyme systems which require these metals for their activity.

Slide 25

Effect of Toxins on Plant Tissue

4. Interfere with the growth regulatory system

Slide 26

Interference of Toxins

Cause an imbalance in the hormonal system and bring about abnormal growth responses

Overgrowths, rosetting, excessive root branching, stem malformation, leaf epinasty, defoliation, and suppression of bud growth.

Slide 27

[Image not found: /uploads/1.2 Toxin_s27_img1.jpg]

Slide 28

Plant growth regulator GA:

Identified as a fungal toxin

First identified in Japan in 1926, as a metabolic by product of the plant pathogen *Gibberella fujikuroi*, which affect rice.

Slide 29

Effect of Toxins on Plant Tissue

5. Inhibit Root Growth

Fusarium moniliforme produces a thermostable toxin even in soil around the roots which causes browning of roots and restrict their development.

[Image not found: /uploads/1.2 Toxin_s29_img1.jpg]

[Image not found: /uploads/1.2 Toxin_s29_img2.jpg]

Figure: Wilt Disease of Soyabean.

Besides,

Stomatal Dysfunction has also been reported for certain pathotoxins

Physical blocking effect of large molecules is also a mechanism of phytotoxicity by toxins.

Slide 30

Causes of toxin in host plant

Toxins are microbial metabolites involved in development of plant diseases.

Toxins are the weapon of microbes, that helps in injuring and killing the host cell.

They cause severe damage to the plant and produce different symptoms.

It inhibits enzymatic reactions of the host.

It is effective at low concentration and directly act upon the host protoplast.

<div id="1-3-effects-of-pathogens-on-plant-physiological-functions"></div>

Lecture: 1.3 Effects of Pathogens on Plant Physiological Functions

Course Slides: 1.3 Effects of Pathogens on Plant Physiological Functions

Slide 1

Effects of pathogens on plant physiological functions

Slide 2

Plant Physiology:

Plant physiology is the study of plant processes

It includes many processes like :

Photosynthesis

Respiration

translocation of water, minerals

Transpiration

[Image not found: /uploads/1.3 Effects of Pathogens on Plant Physiological Functions_s2_img1.jpg]

Slide 3

Photosynthesis

Photosynthesis is the basic function of green plants:

It enables them to transform light energy into chemical energy, which they can utilize in all cell activities.

Photosynthesis is the ultimate source of nearly all energy used in all living cells, plant or animal

Slide 4

Q

Effects of pathogens on photosynthesis

How do we tell that photosynthesis has been affected?

Pathogens damage leaf tissues – reduced photosynthetic area examples of such diseases

Some diseases destroy chloroplasts and cause their disintegration

Pathogens produce toxins such as tentoxin and tabtoxin which inhibit enzymes involved in photosynthesis

Vascular pathogens reduce chlorophyll content and photosynthesis stops before wilting

Viruses and mollicute cause chlorosis and stunting – photosynthesis is greatly reduced

Slide 5

[Image not found: /uploads/1.3 Effects of Pathogens on Plant Physiological Functions_s5_img1.jpg]

[Image not found: /uploads/1.3 Effects of Pathogens on Plant Physiological Functions_s5_img2.jpg]

Spot with necrosis

Blight

Slide 6

Effect on translocation of water and nutrients uptake

Vascular pathogens affect movement of water from the roots

Vascular pathogens also affect movement of nutrients from soil to the leaves – rest of plant deprived of nutrients

If water to leaves is inhibited, leaves cannot function properly – photosynthesis reduced or stopped

Some pathogens affect the integrity or function of the roots

Other interfere with water economy by causing excessive transpiration through leaves and stomata

Q

Slide 7

Diseases caused by Oomycetes Seed rots and Seedling Damping-off

[Image not found: /uploads/1.3 Effects of Pathogens on Plant Physiological Functions_s7_img1.png]

Slide 8

[Image not found: /uploads/1.3 Effects of Pathogens on Plant Physiological Functions_s8_img1.jpg]

Damping-off Death of a seedling before or shortly after emergence due to decomposition of the root and/or lower stem; it is common to distinguish between preemergence damping-off and postemergence damping-off (Pythium damping-off of verbena)

Slide 9

Q

Interfere with translocation in xylem vessels

Physical presence (mycelium, conidia, bacterial cells) in xylem

Polysaccharides in the vessels

Collapse of vessels

Development of tyloses.

Reduced water tension in vessels due to pathogen-induced alteration in foliar transpiration

Slide 10

[Image not found: /uploads/1.3 Effects of Pathogens on Plant Physiological Functions_s10_img1.jpg]

Bean root rot

Slide 11

[Image not found: /uploads/1.3 Effects of Pathogens on Plant Physiological Functions_s11_img1.png]

[Image not found: /uploads/1.3 Effects of Pathogens on Plant Physiological Functions_s11_img2.png]

Wilt: Drooping of leaves and stems from lack of water (inadequate water supply or excessive transpiration); vascular disease that interrupts normal water uptake.

Slide 12

Root and Crown Rot

[Image not found: /uploads/1.3 Effects of Pathogens on Plant Physiological Functions_s12_img1.jpg]

Slide 13

[Image not found: /uploads/1.3 Effects of Pathogens on Plant Physiological Functions_s13_img1.jpg]

Canker The death of cambium tissue by the formation of sharply delineated, dry, necrotic, localized lesions on the stem

Slide 14

Q

Translocation of nutrients through phloem

Pathogens may interfere with movement of nutrients from leaf cells to phloem.

Rusts and mildews cause accumulation of photosynthetic products in areas invaded by pathogens

Infected areas characterized by reduced photosynthesis and increased respiration

Canker causes pathogens to attack the stem and remain confined there.

Pathogen may attack and destroy phloem elements – interfere with downward translocation of nutrients.

Slide 15

Dieback:

Progressive death of shoots and twigs generally starting at the tip of the infected plant part

[Image not found: /uploads/1.3 Effects of Pathogens on Plant Physiological Functions_s15_img1.jpg]

Slide 16

[Image not found: /uploads/1.3 Effects of Pathogens on Plant Physiological Functions_s16_img1.jpg]

Downy mildew: A plant disease in which the fungus appears as a downy growth on the host surface; caused by a member of the Oomycetes.

Slide 17

[Image not found: /uploads/1.3 Effects of Pathogens on Plant Physiological Functions_s17_img1.jpg]

powdery mildew

powdery mildew

powdery mildew

powdery mildew

Powdery mildew

White, powdery, superficial ascomycetous fungus that is an obligate parasite and generally is found only on one or a few closely related species of plants

Slide 18

[Image not found: /uploads/1.3 Effects of Pathogens on Plant Physiological Functions_s18_img1.jpg]

Rust: A disease caused by a specialized group of Basidiomycetes that often produces spores of a rusty color (stem rust of wheat)

Slide 19

Effect of pathogens on host plant respiration

Rate of respiration higher in infected plants

Infected tissues use up reserve carbohydrates faster than healthy tissues

Respirations higher in infected resistant cultivars – higher energy required for production and mobilization of defense mechanisms

Increased respiration has been noticed in cereal rusts, powdery mildews, blast of rice, late blight of potato and many other diseases.

Level of many enzymes associated with respiratory processes is increased in diseased plants.

Diseased plants carry out more fermentation than the healthy plants.

Slide 20

[Image not found: /uploads/1.3 Effects of Pathogens on Plant Physiological Functions_s20_img1.jpg]

Slide 21

Effect on permeability of cell membranes

Cell membrane permeability is altered

Uncontrolled loss of electrolytes, water soluble ions

Inflow of excess undesirable substances

Slide 22

Effect on Transpiration

Transpiration is the process of water movement through the plant and its evaporation from aerial parts, such as leaves, stem and flowers.

It is normally observed that the rate of transpiration increases in plants in which the plant diseases infect the leaves (e.g., leaf-rusts, apple scab, powdery mildews, and downy mildews).

It happens because these diseases result in destruction of the part of the protection devices like cuticle, an increase in permeability of leaf cells, and the dysfunction of stomata.

Slide 23

Transpiration

Increases when plant pathogens infect leaves because of an increase in leaf cell permeability and stomata dysfunction.

Destruction of a considerable portion of the cuticle and epidermis results in an uncontrolled loss of water from the affected areas.

If water absorption and translocation cannot keep up with excessive loss of water, loss of turgor and wilting of leaves.

Slide 24

Effects of pathogens on transpiration

Disintegration of cuticle.

Collapse of vessels.

Formation of tyloses.

Powdery mildew- Epidermal cells invaded by the fungus.

Rust- Leaf surface get exposed due to rupture of epidermis cause unrestricted loss of water.

Blight- Number of active and healthy cells reduced, transport of water to leaves by xylem is reduced.

<div id="2-1-dissemination-of-pathogens"></div>

Lecture: 2.1 Dissemination of Pathogens

Course Slides: 2.1 Dessimation of the Pathogen

Slide 1

Q

Pathogen:

An entity that can incite disease.

Inoculum:

The pathogen or its parts that can cause infection; that portion of individual pathogens that are brought into contact with the host.

Slide 2

Kinds of inoculum produced by pathogens

Plant pathogenic organisms produce several kinds of inocula-

Fungi:

In case of fungi different kinds of inocula are produced-

Different kinds of spores/Conidia: Ascospore,

Basidiospore, Teliospore, Pycnidiospore

Slide 3

The growing mycelium. e.g. soil-borne pathogens

The dormant mycelium in seeds or in often plant parts e.g. late blight pathogen in tubers, loose smut pathogen in seeds

Mycelial strands or rhizomorph

Sclerotia: Sclerotia are of various sizes and shapes. e.g. *Sclerotium rolfsii*, *Rhizoctonia solani*

Slide 4

Virus and Bacteria

No special inoculum is produced by virus and bacteria (except the genus *Streptomyces*, it produces spores). The individual virus particle and the individual bacterial cell constitute the inoculum.

Nematodes

In case of nematodes either the organism itself or its egg act as inoculum.

Parasitic flowering plants

Seeds of parasitic higher plants act as inoculum.

Slide 5

Dissemination/dispersal/transmission

transport of spores or infectious bodies, acting as inoculum, from one host to another host at various distances resulting in the spread of the disease

Displacement of a plant pathogen from its place of production or origin to a suitable place where it can grow/establish

Slide 6

EFFECTIVE DISPERSAL

If displacement occurs on a susceptible host and where the environment is suitable, the pathogen can grow and multiply

Ineffective Dispersal

When plant pathogen does not get any medium for its survival. For example, if it falls on a stone

Slide 7

The dispersal of infectious plant pathogens in space occurs through two ways:

Autonomous or direct or active dispersal

In this method the dispersal of plant pathogens takes place through soil, seed and planting material during normal agronomic operations. There is no major role of external agencies like insects, wind, water, etc. in this type of dispersal.

Indirect or passive dispersal.

in this type of dispersal insects, wind ,water, Human etc.

Slide 8

Q

What is dissemination?

Displacement of a plant pathogen from its place of production/origin to a suitable place where it can grow/establish.

Transport of spores or infectious bodies, acting as inoculum, from one host to another host at various distances resulting in the spread of the disease.

Slide 9

Q

Dissemination

Ineffective

When the displaced plant pathogen doesn't get any medium for its survival.

Effective

When the displacement occurs on susceptible host and where the environment is suitable for the displaced pathogen to establish infection.

Slide 10

Q

Why Pathogen Disseminate (Importance)

Food and nutrition

Survival

Complete life cycle

Reproduction

Slide 11

Types of dissemination of plant pathogens

Indirect Dissemination

Direct Dissemination

through seed or other planting materials.

External

The spores of *Ustilago hordei* are transmitted through adherence to the seed surface

Internal

Passive

By means of external agents

Active/ autonomous dispersal

By growth or mobility of pathogens

Vegetative

Phytophthora infestans causes late blight of potato

Germinative

Ustilago tritici (Loose smut of wheat)

Slide 12

Q

Agents of dissemination

Wind

Water

Soil

Seed

Animal

Human

Machinery used in agriculture

Transport system

Slide 13

Wind dissemination

Slide 14

Characteristics of Wind disseminating fungi

Numerous spore

discharge of spores with sufficient force

Light in weight

smaller size

Thick outer wall

Less affinity to water

E.g. Most fungi except synchytriaceae, Plasmodiophoromyces, Fusarium

Bacteria *X. malvacearum*, *Erwinia amylovora*

Nematode *Anguina tritici*, *Heterodera*

Slide 15

Wind as a means of dispersal

Short distance dissemination

sporangia of downy mildew fungi, conidia of powdery mildew fungi and basidiospores of rust fungi

Long distance

Uredospores of rust fungi, Chlamydospores of smut fungi and conidia of *Alternaria*, *Helminthosporium* and *Pyricularia*

Slide 16

Factors affecting wind dissemination

Periodicity of wind

Diurnal

Nocturnal

Air current

Steady-horizontal

Breeze & turbulence-Vertical

Irregular-Multidimensional

Slide 17

Some Example

P. Graminis tritici

Horizontal---600Km

Altitude-----500m above of the infected field

P. Infestans- Horizontal---200m to 600Km

Spores of Molds

Aspergillus

Penicillium

Cladosporium

Altitude-----2000m above

Slide 18

Dispersal by water

Fungi dissemination

Surface running water

after heavy rains

during irrigation: canals and wells

Ex: The mycelial fragments, spores or sclerotia of fungi, *Colletotrichum falcatum*, *Fusarium*, *Ganoderma*, *Macrophomina*, *Pythium*, *Phytophthora*, *Sclerotium*, etc.

Long distance ----- floods

Slide 19

By rain splash

Most efficient means for bacteria.

Rain drops falling with force on sori, pustules, cankers or even soil surface may splash the propagules

---in small droplets and enable them to land on neighboring healthy susceptible surfaces or

---the water droplets may be carried to long distances by air.

Slide 20

Ex: Bacterial leaf spot of rice (*Xanthomonas campestris* pv. *oryzae*), Bacterial leaf streak of rice (*Xanthomonas campestris* pv. *oryzicola*)

Fungal spores and bacteria present in the air or plant surface are washed downward by rain splash or drops from overhead irrigation and are deposited on susceptible healthy plants.

Slide 21

Insects dissemination

Fungal pathogens

honey secretions having attractive odours. Ex: Sugary disease of sorghum, ergot of rai

Bacterial pathogens

Cucumber wilt- E. tracheiphila -----stripped cucumber beetles (*Acalymma vittata*)

Slide 22

Viral pathogens

More than 80 % of viral and phytoplasma diseases

Largest family- Homoptera- Aphids and leaf hoppers

mealy bugs and scale insects (Coccoidae)

whiteflies (Aleyrodidae)

hoppers (Membracidae) in Homoptera also transmit virus diseases

Slide 23

Insects dissemination

Mechanical

Fungi-----Legs

Bacteria-----Bristles

Viruses-----Wings

Mycoplasma-----Abdomen

Slide 24

Specificity of Insect Transmission

Aphids-----Mosaic group-----Tomato mosaic, Papaya mosaic, potato leaf roll, chili mosaic

Leaf Hopper-----Yellows -----Tungro of rice

Thrips---Spots with concentric ring---Pineapple yellow spots

Whiteflies-----Thickening of veins/veinlets-----Tobacco leaf curl, Tomato leaf curl

Slide 25

Dispersal of plant pathogens by insects

Viruses

different insects

sucking and chewing or biting type mouth parts

Hemiptera.

two families:

Aphidae (aphids)

Cicadellidae (Leaf hoppers) .

A few belong to the aleyroidae (whiteflies) and the coccidae (mealy bugs).

Slide 26

More than fifty species of aphids are known to transmit the viruses of plants.

Myzus persicae alone can transmit more than fifty species.

Aphids: bean common mosaic virus, bunchy top mycoplasma, potato leaf roll virus, chilli and tomato mosaic.

White flies: leaf curl of tomato and papaya. Tungro virus is transmitted by leaf hoppers and brown plant hoppers.

Slide 27

The transmission of plant viruses by insects may be by two ways:

Mechanical :

When the viruses are simply carried externally or internally by the insects.

Insects have biting type of mouth parts e.g. coleoptera beetles

squash mosaic,

cowpea mosaic,

turnip yellow mosaic

Slide 28

2. Biological: When the pathogens are transmitted through sexual process from generation to generation. Insects having sucking type of mouth parts such as aphids, leafhoppers, whiteflies, mealy bugs.

Slide 29

Biological transmission are of three types:

a. Non-persistent (or) stylet-borne: Viruses requiring no incubation time e.g. Potato virus Y

b. Semi-persistent (or) foregut-borne

c. Persistent : Viruses requiring incubation time

e .g. Potato leaf roll virus

- circulative (no virus replication)
- circulative-propagative (virus replication)

Slide 30

Non- and Semi-persistent transmission

- virus is not internalized in the vector
- virus do not enter the hemocoel and salivary glands of the vector

Persistent circulative transmission

- virus is internalized in the vector

- virus enters the hemocoel and salivary glands of the vector

Slide 31

Persistent circulative-propagative transmission

- virus is internalized in the vector
- virus enters the hemocoel and salivary glands of the vector
- virus multiplies in the insect vector

Slide 32

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Slide 33

Fungi transmitted by insects:

About 66 species of fungi causing plant diseases are transmitted by more than a 100 species of insects belonging to at least six orders.

Transmission occurs in various ways with or without injury to the plant.

Slide 34

For example, spores of *Ustilago* spp and *Botrytis* spp are carried from the infected anthers to the uninfected flowers by pollinating insects such as Bees, wasps.

Aphids can carry the mycelial fragments and sporangia of *Phytophthora infestans* on their bodies and leg for the short distance dissemination.

Slide 35

Bacteria transmitted by insects:

For example, Cucumber beetle chew the plant parts infected by cucurbit wilt pathogen, *Erwinia tracheiphila* and carry the bacterium inside the body and passing on the host in the next season

Seed corm maggot can disseminate the black leg pathogen, *Pectobacterium carotovora* susp *carotovora*.

Xanthomonas axonopodis pv. *citri* (citrus canker pathogen) is disseminated by citrus leaf miner.

Mycoplasma transmitted by insects:

Such as Psyllid bug transmits citrus greening mycoplasma and whitefly transmits little leaf mycoplasma.

Slide 36

Dispersal or transmission of plant pathogens by Seed

Most effective and the easiest way of transmission

Approximately 500 plant species are attacked by seed-borne pathogens.

Each of the cereal crops rice, wheat, sorghum and maize may be attacked by nearly 500 different seed-borne pathogens.

It appears that as many as 41 seed-borne pathogens occur on bean and 3 dozens of different pathogens are transmitted through soybean seeds.

Slide 37

Example:

Many seed-borne pathogens have worldwide distribution and under favourable conditions they may cause disease in epidemic form.

Fungi

Loose smut of wheat (*Ustilago tritici*)

Leaf blight of wheat (*Bipolaris sorokiniana*)

Blast of rice (*Pyricularia oryzae*)

Brown spot of rice (*Bipolaris oryzae*)

Slide 38

Bacteria

Erwinia carotovora (Soft rot of potato)

Xanthomonas oryzae pv. *oryzae* (Leaf blight of rice)

Common bacterial blight of bean (*Xanthomonas phaseoli*)

Nematodes

Ditylenchus angustus (Ufra of rice)

Aphelenchoides besseyi (White tip of rice)

Virus

Bean common mosaic virus

Papaya leaf curl virus

Slide 39

Mechanism of seed transmission:

Pathogens may exist with the seeds in two ways:

Infection

Contamination

Types of association	Parts infected/ contaminated	Category of infection/ contamination
Seed infection	Embryo	Embryo infection
	Endosperm, seed coat/ pericarp	Extra-embryal infection

Seed contamination	Seed coat/ Pericarp	Contamination
	Bract, glume etc.	

Slide 40

Dispersal of plant pathogens by nematodes

Some viruses as well as some bacterial and fungal plant pathogens are transmitted by nematodes.

Examples:

Corynebacterium fascians is disseminated by ectoparasitic nematodes, *Aphelenchoides*

Yellow ear rot pathogen of wheat, *Corynebacterium tritici* is dispersed by the ear cockle nematode, *Anguina tritici*.

Slide 41

Fungi

There is evidence that some root infecting pathogens, particularly fungi, may enter roots through wounds made by nematodes. Such as cotton wilt caused by *Fusarium oxysporum* f. sp. *vasinfectum*, the pathogen being associated with root knots.

Slide 42

Bacteria

Bacterial wilt caused by *Ralstonia solanacearum* is more severe with the presence of root knot nematodes which probably facilitate the entry of the bacteria thus help in disseminating the pathogens.

Slide 43

Viruses

The grapevine fan leaf virus is disseminated by *Xiphinema* index. Some polyhedral and tubular viruses have also been shown to be nematode transmitted, the former by the species of *Xiphinema*, *Longidorus* and *Trichodorus* spp.

Slide 44

Dispersal of plant pathogens by human

one of the important agents.

short and long distance.

day to day activities

Through propagating materials

propagation

field to field, orchard to orchard and from one geographic region to another.

Slide 45

ii. Through import and export of seeds and other propagating materials

Xanthomonas oryzae pv. *oryzicola* (bacterial leaf streak pathogen),

potato viruses,

Ustilago nuda (loose smut of wheat),

Sphacelotheca sorghi (grain smut of jawar),

Neovossia indica (kernal bunt of wheat)

Anguina tritici (ear cockle of wheat).

Slide 46

iii. Movement of farm workers: Spores and other vegetative and reproduction structures (sclerotia etc) can be carried by worker clothings, shoes, hands etc. from plant to plant and from field to field.

Slide 47

Examples: Angular leaf spot of cucurbits caused by *Pseudomonas lachrymans*. When cucumbers are grown for pickles, the fruits are picked. When a person comes in contact with or picks an infected plants, his hands become contaminated , particularly when the plants are wet with dew. The pathogen, is thus, spread down the row of the plants.

Slide 48

The virus that are spread by animals can also be spread by man. Tobacco Mosaic Virus is known to survive in dry tobacco leaves. Hence, workers who handle tobacco plants in the field are advised not to smoke while they are working and to thoroughly wash their hands before entering the field.

Slide 49

iv. Use of contaminated implements: cutting knives etc. helps in transmission of soil-borne pathogens and also the pathogens present in bulbs, tubers, sugarcane setts etc.

v. Through grafting and budding between healthy and diseased plants: It is the most effective method of distribution of pathogens of fruit trees such as the careless selection of stock and scion for propagating citrus trees.

Slide 50

vi. Cultural operations:

Hoeing, weeding, pruning carried out by man not only transmit diseases by contact between the plant and contaminated tools but also by providing wounds on plant surface for entry of pathogens that have no capacity to cause direct penetration

Slide 51

Dispersal by phanerogamic parasites:

Phanerogamic parasites transmit the viruses by acting as a bridge between the diseased and healthy plants.

Ex: Dodder

(*Cuscuta California*, *C. campestris*, *C. subinclusa* etc.)

Cuscuta subinclusa ---- Cucumber mosaic virus

Cuscuta californica ----- Tobacco mosaic virus, Tobacco rattle virus, Tomato spotted wilt virus

Cuscuta campestris - Tomato bushy stunt virus

Slide 52

Dispersal by animals:

Cattle feeding on diseased fodder ingest the viable fungal propagules (spores or oospores or sclerotia) and pass out as such in the dung.

Soil inhabiting fungi especially sclerotia adhere to the hoofs and legs of animals and get transported to other places.

Slide 53

Dispersal by other mammals and birds:

Pathogens that can persist on clothing or machinery might easily be carried on an animal's hair or a bird's feathers.

Examples:

Rabbits and dogs spread potato virus X in potato crops when they brush past infected plants and break the leaf hairs.

Birds carry the chestnut blight fungus (*Endothia parasitica*). Birds also spread the seeds of parasitic higher plants (dodder)

<div id="2-2-epidemiology-of-plant-diseases"></div>

Lecture: 2.2 Epidemiology of Plant Diseases

Course Slides: 2.2 Epidemiology

Slide 1

What is Plant Pathology or Phytopathology?

Pathology comes from 2 Greek words

Pathos

Suffering

Logos

Study

Plant Pathology or Phytopathology is the study of suffering of plant diseases.

Slide 2

Why do we study Plant Pathology?

To study the causes of diseases

To study the interaction(s) between the plant and the pathogen

To study the mechanism(s) of disease development.

To develop management strategy of the diseases

Slide 3

Disease = Disturbance from plant pathogen or environmental factor that interferes with plant physiology

Causes changes in plant appearance or yield loss

Disease results from:

Direct damage to cells

Toxins, growth regulators, or other byproducts that affect metabolism

Use of nutrients and water or interference with their uptake

Slide 4

Epidemic Gr. Epi=upon, among and Demons=people Epidemic What is among people

"Change in disease intensity in a host population over time and space."

Change: often increase -- a dynamic process

Disease: dealing with diseases, not just the pathogen (or plant/crop)

Host: Organism infected (or potentially infected) by another organism

Population: a population phenomenon

Time and space: two physical dimensions of interest.

Slide 5

Q

Epidemic and Epidemiology

When a pathogen spreads to and affects many individuals within a population over a relatively large area and within a relatively short time, the phenomenon is called an epidemic.

An epidemic has been defined as any increase of disease in a population.

A similar definition of an epidemic is the dynamics of change in plant disease in time and space.

Slide 6

Plant Disease Epidemiology

Study of epidemics

The study of epidemics and of the factors that influence them is called epidemiology.

Epidemiology is concerned simultaneously with populations of pathogens and host plants as they occur in an evolving environment and human interferences.

Epidemiology or epiphytology is the study of the outbreak of disease, its course, intensity, cause, effects and the various factors governing it.

Slide 7

History (ancient to modern times)

Hippocrates (~400 BC): First use of "epidemic", widespread disease (human diseases)

Theophrastus (~340 BC): Plant diseases in fields, Environmental influences

Pliny (~50 AD): Plant diseases; soil; climate

Duhamel de Monceau (1728 AD): Disease progress curves, Comparison of plant and animal epidemics

Late 19th Century and forward...0

Kuhn (1858) - 1st textbook of plant pathology

Ward (1901): book "Diseases in Plants" emphasized ecology (populations) of disease

Jones (1913) - role of the environment

Gaumann (1946): "Principles of Plant Infection" -Disease spread, -Conditions leading to an epidemic, -'Infection Chain' (= disease cycle), -compare with medicine (diseases of humans)

Disease

7

Slide 8

Vanderplank (1963) (used to be van der Plank)

"Plant Diseases: Epidemics and Control"

-Populations

- Rates (dynamic processes)
- Analysis, mathematics
- Models, theory
- Link epidemiology and control
- Established the science of plant disease epidemiology

Other pioneers:

Zadoks (1960-1995), The Netherlands

Kranz (1968-1995), Germany

Waggoner (1960-mid --1980s), USA

S. Nagaranjan 1983-India

Note: many developments in other fields...

Ecology, medical epidemiology,

Biomathematics, etc.

Large (1952, and others)

- Disease progress curves

- Crop losses

- Disease assessment (measurement)

Horsfall & Dimond (1960)- "Plant Pathology, Volume 3"

- Populations

- Inoculum density: disease relations

- Spore dispersal

- Analysis (mathematics)

- Forecasting, prediction

- Traditional definition ---> Modern definition

Gregory (1963, 1973)

"The Microbiology of the Atmosphere"

-spore dispersal, disease spread

Aerobiology

8

Slide 9

Q

Based on the occurrence and geographical distribution they are classified as follows:

1. Endemic or Enphytotic When a disease is more or less constantly occurring year after year in a moderate to severe form in a country or locality then it is called as endemic disease. eg: wart disease of potato (*Synchytrium endobioticum*) is endemic in Darjeeling, citrus canker (*Xanthomonas axonopodis* pv *citri*) in Asia and sorghum rust (*Puccinia purpurea*)

Certain disease are endemic in one area and become epidemic in another area. Eg: Citrus canker is endemic in Asia but epidemic in the introduced place, Florida (U.S.A). The downy mildew of corn is a endemic disease in India but became epidemic in the Philippine

Slide 10

Q

2. Epidemic or Epiphytotic

It is a sudden outbreak of a disease periodically over a widespread area in a devastatingly severe form causing severe losses or complete destruction.

This is constantly present in a locality but it assumes severe form only on occasions.

This is because of the occurrence of favorable environment responsible for the rapid development of disease.

eg: wheat stem rust (*Puccinia graminis tritici*), late blight of potato (*Phytophthora infestans*) Irish famine 1845-1846, and rice blast (*Pyricularia oryzae*) Bengal famine 1943.

Slide 11

Q

3. Pandemic

When an epidemic disease spreads over continents or subcontinents and involves mass mortality it is considered as pandemic. The outbreak of black stem rust of wheat in India during 1947 is best example for a pandemic disease.

4. Sporadic

Diseases which occur at irregular intervals over limited areas or locations are called sporadic. They occur relatively in few instances. Eg: Fusarium wilt of cotton (*Fusarium oxysporum* f sp. *vasiinfecum*) and loose smut of wheat (*Ustilago tritici*).

Slide 12

Q

Vanderplank's Equivalence Theorem

"Effects of host, pathogen and environment can be translated into terms of the rate parameter of an epidemic"

Changes in any component has an equivalent effect on disease

-More-less susceptible host

-More-less favorable environment

-More-less amount of aggressive pathogen disease

Slide 13

Q

Therefore, disease management principles and practices are often centered around the concept of the Disease Triangle. so that management tactics often seek to manipulate one or more of the components of the disease triangle.

Slide 14

Elements of an epidemics

An epidemic may cause widespread and mass destruction of crop in a short time or may persist for long periods depending upon the following factors responsible for the disease:

Slide 15

Q

Elements of a disease cycle

Disease development is also affected by

4. Time
5. Humans
- 1.Host
2. Pathogen
3. Environment

Interactions of the 3 main components are described by the disease triangle.

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[Image not found: /uploads/2.2 Epidemiology_s15_img2.png]

Interactions of the 5 components are

described by the disease pyramid.

The Disease Triangle

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The Disease Triangle

[Image not found: /uploads/2.2 Epidemiology_s16_img1.png]

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[Image not found: /uploads/2.2 Epidemiology_s17_img1.png]

Temp. Fluctuation Effects

Slide 18

Factors that affect the development of epidemic

Slide 19

Host factors

A. Level of genetic resistance or Susceptibility of the host

Susceptibility- The lack of ability to resist the effect of a pathogen or other damaging factors.

Resistance- The ability of an organism to exclude or overcome pathogens completely or in some degree.

a. true resistance-gene control, eg: vertical and horizontal resistance

b. apparent resistance- not genetically control

disease escape- to overcome the disease developing factors

disease tolerance- disease occur but not economic loss

Race : A genetically and often geographically distinct mating group within a species of a pathogen.

Q

Slide 20

Host factors

B. Degree of genetic uniformity of the host plant

Eg : Vegetative propagated crop, self-pollinated crop, monoculture, especially clones are more susceptible to diseases.

Q

C. Types of crop-

- Annual crops & foliar or fruit diseases develop much more rapidly (in few weeks)
- Perennial woody diseases take longer time to develop

(in years).

D. Age of crop plant –

Plant change in their susceptibility to disease with age. Some plants are susceptible only during growth period & become resistant during mature period.

Q

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[Image not found: /uploads/2.2 Epidemiology_s21_img1.jpg]

E. Introduction of new hosts-

Slide 22

Pathogenic Factors

Level of virulence and aggressiveness –

Virulence- Degree of pathogenicity of a given pathogen.

Aggressiveness – Production of a large number of inoculums within a very short period of time.

B. Quantity of inoculums near host-

More quantity, more probability of causing diseases.

Q

Slide 23

C. Types of reproduction of the pathogen-

A central concept to epidemiology is that different pathogen populations have different disease cycles.

I. Monocyclic = single cycle (simple interest)

Pathogens that complete one or even part of one disease cycle/year are called monocyclic

In monocyclic pathogens the primary inoculum is the only inoculum available for the entire season. There is no secondary inoculum and no secondary infection.

The amount of inoculum produced at the end of the season, however, is greater than at the start of the season so the amount of inoculum may increase steadily from year to year.

Eg. Loose smut of wheat

Q

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Loose smut of wheat

Slide 25

This representation of plant disease over time is referred to as a “Disease Progress Curve”

Graphically, disease caused by monocyclic pathogens looks like a saturation curve.

[Image not found: /uploads/2.2 Epidemiology_s25_img1.jpg]

Slide 26

Basis of modelling :

All the epidemiological models are based on the cyclic pattern of disease occurrence.

Monocyclic disease

Polycyclic disease

How to reduce disease incidence, x , at any point in the epidemic: Reduce initial inoculum, Q (monocyclic) x_0 (Polycyclic).

Reduce rate constant, R .

Reduce duration of epidemic, t .

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The monocyclic model (Van der plank in 1963)

It is clear from the above model of a monocyclic epidemic that Q , R , and t have equal weight in their effect on x . A reduction in the initial inoculum or the rate of infection will result in a reduction in the level of disease by the same proportion at any time(t), throughout the epidemic. If t can be reduced (for example, by shortening the season), disease will be reduced proportionately.

Slide 28

Examining these models, we can see that in both there are three ways in which we can reduce x at any point in the epidemic:

Reduce the initial inoculum (Q in the monocyclic model and x_0 in the polycyclic model). (Actually x_0 is the initial incidence of disease, which is proportional to the initial inoculum)

Reduce the rate of infection (R in the monocyclic model and r in the polycyclic model)

Reduce the duration of the epidemic (the time, t , at the end of the epidemic)

Slide 29

Effect of x_0 on epidemic development:

x_0 depends upon:

- Inoculum from previous crops within a field.

- Inoculum from crops in adjacent fields.

xo is affected by:

- By destroying infected plant parts.

- By removing diseased plants.

- Through chemical seed treatments.

- Use of protective fungicides.

- Race-specific disease resistance.

- Biological control agents targeted at initial inoculum.

Slide 30

Effect of r on epidemic development:

r depends upon:

- Reproductive potential of the pathogen.

- Virulence of the pathogen.

- Susceptibility of the host.

- Favorability of environment.

r is affected by:

- Non-specific disease resistance.

- Use of systemic fungicides.

- Cultural practices that alter environment.

- By removal of diseased plants.

Slide 31

Rate of increase of disease over time can be represented by a simple interest function

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[Image not found: /uploads/2.2 Epidemiology_s31_img2.wmf]

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Examples of Monocyclic Diseases

Blackleg of potato (*Erwinia caratovora*)

Verticillium wilt

Cereal Cyst Nematode

Slide 33

Q

Polycyclic = multiple cycles/year (compound interest)

The pathogens that go through more than one (2-30) disease cycle in a growing season and are referred to as polycyclic.

Only a small number of sexual spores or other hardy structures survive as primary inoculum that cause initial infections.

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Once infection takes place, large numbers of asexual spores are produced as secondary inoculum at each infection site.

These spores can produce new (secondary) infections that produce more asexual spores and so on.

With each cycle the amount of inoculum is multiplied manyfold.

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Graphically this type of population growth is represented as a sigmoid curve

[Image not found: /uploads/2.2 Epidemiology_s35_img1.jpg]

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[Image not found: /uploads/2.2 Epidemiology_s36_img1.jpg]

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The polycyclic model

If r is very high, the apparent

effect of reducing x_0 is to delay

the epidemic.

If r is very high, x_0 must be reduced to very low levels to have a significant effect on the epidemic.

Reducing r has a relatively greater effect on the epidemic than reducing x_0 .

Reducing x_0 makes good strategic sense only if r is low or if r is also being reduced.

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[Image not found: /uploads/2.2 Epidemiology_s38_img1.jpg]

Rate of increase of disease over time can be represented by a compound interest function.

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Many of these polycyclic pathogens are disseminated primarily by air or air-borne vectors and are responsible for most of the explosive epidemics in most crops

Examples of Polycyclic Diseases

Downy mildews

Powdery mildews

Late blight of potato

Leaf spots

Blights

Grain rusts

Aphid borne viruses

Root-knot nematodes

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[Image not found: /uploads/2.2 Epidemiology_s40_img1.jpg]

Q

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[Image not found: /uploads/2.2 Epidemiology_s41_img1.jpg]

Q

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III. Polyetic (multi-year) cycles

The pathogens that take several years before the inoculum they produce can be disseminated and initiate new infections are called polyetic pathogens.

May not cause many new infections over a given area in a year, amount of inoculum does not increase greatly within a year.

However, because they survive in perennial hosts they have almost as much inoculum as they had at the end of the previous year.

Inoculum may increase steadily(exponentially) from year to year and can cause severe outbreaks when considered over several years.

Slide 43

Examples of Polyetic Diseases:

1. Some diseases of trees:

Dutch elm disease

Citrus tristeza

Fungal vascular wilts

2. Mycoplasmal yellows
3. Viral infections

Slide 44

Pathogenic Factors

D. Ecology of the pathogen-

Presence of inoculums in the host

Vascular fungi reproduce inside of the host and vector transmitted cause epidemic

Soil borne pathogen produce inoculums on the infected field.

E. Mode of spread of the pathogen-

- Wind, Water, Soil, Seed, insect, Human, Machinery used in agriculture

Slide 45

Q

Environmental Factors

Temperature

Host effect- High temperature break the host resistance especially vertical resistance.

Pathogen effect- High temperature increase the growth of pathogen but low temp. reduce the amount of inoculum also reduce the movement of vectors.

Slide 46

Q

Environmental Factors

Moisture

Activates resting stages

Affects germination of spores and penetration into host eg. Soil borne pathogen produce inoculums on the infected field

Water on leaves

Humidity and

Splashing water- distributes inoculum

Slide 47

Q

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Moisture

Effect of foliage density on development of *Phytophthora infestans* during a period of partly favorable weather (May–June) and of very favorable weather (November– December).

Slide 48

Human Factors

A. Site selection and preparation

-Poor drainage, low land, infected field.

B. Selection of propagating materials

-Using pathogen free or treated planting materials reduce disease.

C. Cultural practices

-Monoculture, higher level of nitrogenous fertilizer, dense planting, overhead irrigation, poor sanitation, creation injury in plant.

D. Disease control

-Cultural , biological chemical control reduce the epidemic.

Slide 49

Q

Disease development is also affected by Time

Time factors

Season of the year

Duration & frequency of favorable temp. & rains

3. Appearance of vectors, etc.

Slide 50

Q

Eg: Outbreak of Phytophthora wilt of betelvine occurs during rainy season in Bangladesh. This disease once again become destructive during rainy season. This type of epidemic is known as seasonal epidemic or annual epidemic.

Eg: Epidemics caused as a result of introduction of new pathogens in the locality. The well known epidemics of late blight of potato in Europe and blast disease of rice in South East Asia.

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Implications for Disease Management Strategies

Monocyclic Diseases

Reduce the amount of primary inoculum, or affect the efficiency of invasion by the primary inoculum.

Polycyclic Diseases

Reducing the amount of primary inoculum has less impact.

Reducing the rate of increase of the pathogen more beneficial.

Slide 56

Principles of epidemiology indicates that control measures can do this in only two ways.

1. They may reduce (or delay) disease at the beginning of the season (x) or
2. They may decrease the rate of disease development (r) during the growing period.

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Ways to reduce disease (inoculum) at beginning (x_0)

Affects monocyclic and polycyclic diseases

Certified seed

Sanitation

Seed treatments

Quarantine

Host plant resistance

Ways to decrease the rate of disease development (infection rate) (r)

Change the environment

Fertilizer application

Host plant resistance

Ways to change (t)

Harvest early before disease becomes severe.

Plant early (cereal cyst nematode)

Slide 58

PATTERNS OF EPIDEMICS

Interactions among the elements of epidemics, as influenced over time by factors of the environment and by human interference, are expressed in patterns and rates. Two types:

Disease–progress curve

Disease gradient curve

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Pattern and Rate of epidemic

Disease Progress

Disease on plants usually starts out at a low level, a small number of plants affected and a small amount of plant tissue affected, and it becomes of concern to us only when its incidence and severity increases with time.

Progressive epidemic

If the epidemic is considered as increase of diseases in extensity that is progressive epidemic.

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Disease Progress

For example, in Phytophthora blight of pepper seedlings (*Phytophthora capsici*) and Fusarium kernel rot (*Fusarium moniliforme*) of maize, disease progress is roughly linear.

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Phytophthora blight of pepper seedlings

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Linear of disease progress

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Q

Disease Progress Curve (DPC)

Rate of epidemic per unit of time.

Saturation type of curve

Three monocyclic diseases of different epidemic rates.

(B) Sigmoidal curve

Polycyclic disease, such as late blight of potato.

(C) Bimodal curve

Polycyclic disease, such as apple scab, in which the blossoms and the fruit are infected at different, separate times

Slide 63

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Different Amount of initial inoculums different epidemic rate.

A. Saturation type- for monocyclic disease

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[Image not found: /uploads/2.2 Epidemiology_s64_img1.jpg]

Eg. Late blight of potato

Three phase- log, exponential and lag phases.

B. Sigmoid type- for polycyclic disease

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Disease affecting different organs

C. Bimodal type

Slide 66

Disease Gradient Curve (DGC)

The percentage of disease and the scale for distance vary with the type of pathogen or its method of dispersal. Distance is small for soil borne pathogens or vectors and larger for airborne pathogens.

The progress of an epidemic measured in terms of changes in the number of lesions, the amount of diseased tissue, and the number of diseased plants as it spreads over distance, is called disease gradient curve (spatial pattern).

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Slide 67

Rate curves

It refers to the amount of increase of disease per unit time in the plant population.

Rate curves are different types. which are as follows-

Bell shaped Rate curve

Symmetrical Rate curve

Asymmetrical Rate curve

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Eg.

Late blight of potato (Sigmoid Curve).

Bell shaped Rate curve

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[Image not found: /uploads/2.2 Epidemiology_s69_img2.jpg]

Eg. Apple scab (Bimodal Curve), mildew.

Symmetrical Rate curve

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[Image not found: /uploads/2.2 Epidemiology_s70_img2.jpg]

Eg. Alternaria leaf blight

Asymmetrical Rate curve

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Relationships between disease cycles and epidemics

Pathogens that complete their life cycle in one year are called monocyclic

Fungi such as smut produce spores at the end of the season to begin the next season

Root rot and vascular wilts require two seasons and alternate hosts to complete their life cycle

Slide 72

Relationships between disease cycles and epidemics

Some pathogens go through many generations in one year

They are called polycyclic pathogens

They can complete from 2 to 30 cycles in per year

Polycyclic pathogens are usually airborne or airborne vectors and can cause explosive epidemics

Slide 73

Relationships between disease cycles and epidemics

Downy mildew, late blight of potato, powdery mildews, leaf spots and blights, grain rusts and insect-borne viruses cause explosive epidemics

The number of spores etc that survive the season is small but multiply rapidly once conditions are suitable

Slide 74

Relationships between disease cycles and epidemics

In some diseases of trees, fungal vascular wilts, phytoplasma declines, and viral infections the pathogen may not complete the life cycle in one year but may take several years to produce inoculum.

These are called polyetic.

Since they remain in the host, they produce exponentially. (Dutch elm disease)

Slide 75

Comparison of Disease Cycles

	Fungi	Bacteria	Viruses	Nematodes
Survival	Crop residue, Soil, Alternate hosts	Crop residues, Soil, Alternate hosts, Insect vectors	Alternate hosts, Insect vectors	Crop residues, Soil
Dispersal	Wind Rain Insects	Wind Rain Insects	Insects	Tillage Equipment Water run-off
Infection	Directly Wounds Insect feeding	Wounds Insect feeding	Insect feeding	Directly

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Lecture: 2.2 Epidemiology - Core/Important Concepts

Course Document: 2.2 Epidemiology(IMPORTANT)

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<div id="3-1-methods-of-plant-disease-control"></div>

Lecture: 3.1 Methods of Plant Disease Control

Course Slides: 3.1 Method of Plant Disease Control

Slide 1

Methods of Plant Disease Control

Slide 2

What is Control Measures.....

Measures taken to prevent the incidence of disease, reduce the amount of inoculum, and minimize the loss are known as control measures.

Plant disease management is an eminent process in crop production.

A disease can be managed by eliminating interaction between

A susceptible host

A virulent pathogen

In suitable environmental condition

Slide 3

Q

Slide 4

Disease control

There are 6 basic types of control measures:

a. Biological control

b. Cultural control

c. Legislative control

Chemical control

Physical control

Host resistance

Slide 5

Q

PRINCIPLES OF PLANT DISEASE MANAGEMENT

1) AVOIDANCE

2) EXCLUSION

3) ERADICATION

4) PROTECTION

5) RESISTANT VARIETIES

6) THERAPY

Slide 6

Avoidance

Q

Avoiding disease by altering planting time, or planting in areas where inoculum is ineffective due to environmental conditions.

Avoidance can be carried out by:-

Choice of geographical areas.
Choice of time of showing.
Selection of seed and planting material.
Selection of field.
Disease escaping varieties.
Modification of cultural practices.

Slide 7

Q

Exclusion

Preventing the inoculum from entering or establishing in the field.

Seed treatment

Inspection

Certification

Quarantine (Federal or State)

Cleaned farm equipment

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Cleaning farm equipment

Slide 8

Q

Eradication

Aimed at eliminating a pathogen after it is introduced into an area but before it has widely spread.

It can be applied to

Individual plants,

Seed lots,

Fields or regions.

It is generally not effective over large geographic areas.

Slide 9

Q

Eradication can be done by.....

Destroying weeds that are reservoirs of various pathogens or insect vectors of disease,

Biological control of plant pathogen

Crop rotation

Soil treatment

Heat and chemical treatment

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Elimination of potato cull piles

Weed Control

Slide 10

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Crop Rotation

Burning of infected residues crop

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Propane flaming

Soil Treatment

Slide 11

Q

Protection

Preventing infection by creating a chemical toxic barrier between the plants and pathogens.

Chemical treatment

Chemical control of insect vector

Modification of environment

Modification of host nutrition

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Slide 12

Protection can be done by.....

[Image not found: /uploads/3.1 Method of Plant Disease Control_s12_img1.jpg]

[Image not found: /uploads/3.1 Method of Plant Disease Control_s12_img2.jpg]

[Image not found: /uploads/3.1 Method of Plant Disease Control_s12_img3.jpg]

Bananas are covered

with plastic sleeves

Ground sprayers

Raised planting beds

Slide 13

Q

Resistant Varieties

Preventing infection or reducing the effect of infection by managing the host through improvement of resistance in it by genetic manipulation or by chemotherapy.

Selection and hybridization of disease resistance

Mutation of disease

resistance

[Image not found: /uploads/3.1 Method of Plant Disease Control_s13_img1.jpg]

Slide 14

[Image not found: /uploads/3.1 Method of Plant Disease Control_s14_img1.jpg]

Susceptible and resistant plant

[Image not found: /uploads/3.1 Method of Plant Disease Control_s14_img2.jpg]

Susceptible and resistant plant to papaya ringspot virus

Slide 15

Q

Therapy of disease plant

Reducing the severity of disease in an infected individual by chemicals.

Chemotherapy

Tree Surgery

Heat Therapy

Ex: Bordeaux Mixture.

[Image not found: /uploads/3.1 Method of Plant Disease Control_s15_img1.jpg]

Slide 16

Cultural control

Q

Cultural control: Manipulation of the environment to control diseases caused by pathogens.

There are many types of cultural control. Here are a few selected examples:

Crop rotation — rotate crops and varieties over seasons to reduce pathogen inoculum levels.

This is probably the most widely employed control measure in agriculture. example: Rotate soybean with corn to control soybean cyst nematode

Slide 17

Cultural control

Q

Selection of planting date or planting location — Choose a time/place favorable for the host, rather than the pathogen: avoid pathogen or its vector example: (Time) Plant cotton late to control damping-off caused by Pythium (warm soil)

Seeding rate and canopy density — Adjust within-row and between-row spacing to open the canopy and reduce diseases that spread in the humid, protected canopy environment

Slide 18

Cultural control

Q

Irrigation

Pathogens can be spread in irrigation water or favored by wet soils-example: late blight (*Phytophthora*)

Pathogens can be controlled by flooding – for example: *Fusarium* wilt on banana

Control insects and weeds — Insect vectors, viruses, other pathogens, and weeds serve as alternate hosts for pathogens or vectors and increase canopy density

Slide 19

Cultural control

Q

Sanitation Keep the area free of diseased plant material by pruning diseased branches (fireblight), plowing under or burning debris, and washing and sterilizing harvesting and processing equipment (*Rhizopus* soft rot); poor sanitation contributed to the late blight outbreak that caused the Irish famine

Balanced fertilizer application and proper tillage – Balanced fertilizer application is treated as a control measure to kill or inhibit the pathogens in the field.

Slide 20

Role of Trap Crops and Cover Crops in Disease Management

Q

[Image not found: /uploads/3.1 Method of Plant Disease Control_s20_img1.png]

Slide 21

Role of Trap Crops and Cover Crops in Disease Management

Q

[Image not found: /uploads/3.1 Method of Plant Disease Control_s21_img1.png]

Slide 22

[Image not found: /uploads/3.1 Method of Plant Disease Control_s22_img1.jpg]

Slide 23

Legislative control

Quarantine — detention and associated practices for preventing the entry of diseased materials or pathogens into an area; relatively inexpensive; can be at the national or state level

Plant Quarantine Act of Bangladesh, 2011

An Act to make provisions, in the context of international traffic in plants and plant products, for preventing the introduction of insects or pests into, and spread thereof within, Bangladesh and for the matters relating to phytosanitary and other measures incidental and ancillary thereto.

Slide 24

Main activities of plant quarantine

Tracing, identification and control of quarantine pests.

Pest risk analysis, evaluation and determination of quarantine conditions for imported and transit plant shipments.

Inspection, sampling, testing and undertaking quarantine regulations on imported and transit agricultural shipments.

Supervision and undertaking post-entry quarantine regulations on plants which are subject to the regulations.

Inspection, sampling, testing and issuing phytosanitary certificate for exporting plant consignments.

Slide 25

Host Resistance

Slide 26

1. Host resistance - control based on the genes and the resistance mechanisms they control

Van der Plank described two types of resistance (the 1960s; these are the "classics"):

Vertical resistance — resistance that is effective against some, but not all, races of a pathogen; decreases the effective amount of incoming inoculum (avirulent races can't infect), but does not reduce the rate of disease development (virulent races are not affected)

Horizontal resistance — resistance that is effective against all races of the pathogen; decreases the rate of disease development for all races

Slide 27

[Image not found: /uploads/3.1 Method of Plant Disease Control_s27_img1.wmf]

Slide 28

b. Resistance has been defined in many other ways since Van der Plank, including systems based on: epidemiologic effects, number of genes involved, how long the resistance lasts under field conditions; additional terms you should be familiar with are:

tolerance — plants are diseased, but they do not yield less than healthy plants

induced resistance — a normally susceptible plant treated with an avirulent strain of a pathogen gives a resistant reaction when challenged later with a strain that is virulent

Slide 29

Physical Methods

Heat : Elevated temperature (42 to 50oC) usually beneficial, but leads to greater loss of seed germination.

Hot water treatment:

Pre-soaking the seed

Pre-heating at 100C for 1 to 2 min

Hot water soak

Drying

Cooling

Hot air treatment

Refrigeration

Solar heat treatment

Slide 30

Combinations – Mung Bean Seed

Dry heat (50oC, 1 h) no effect on germination and sprout growth rate

Dry heat (50oC, 1 h) followed by gamma irradiation (2.0 kGy)

→ no effect on germination, reduced sprout growth rate

Slide 31

Hot Water – Mung Bean Seed

Hot water

55oC/20 min

60oC/10 min

70oC/5 min

80oC/2 min

None of the treatments reduced seed germination below 95%. Thicker seed coat protects the embryo.

Slide 32

Protocol for Tomato Seed:

Preheat seed for 10 min. at 100°F (37°C).

Transfer to 122°F (50°C) for 25 min.

Cool seed immediately and dry completely prior to sowing or re-packaging. Seed may be treated with fungicidal dust when dry.

Hot-Water Seed Treatment

From S.A. Miller, Ohio St. Univ.

[Image not found: /uploads/3.1 Method of Plant Disease Control_s32_img1.jpg]

[Image not found: /uploads/3.1 Method of Plant Disease Control_s32_img2.jpg]

[Image not found: /uploads/3.1 Method of Plant Disease Control_s32_img3.jpg]

Slide 33

[Image not found: /uploads/3.1 Method of Plant Disease Control_s33_img1.jpg]

Slide 34

[Image not found: /uploads/3.1 Method of Plant Disease Control_s34_img1.png]

Slide 35

[Image not found: /uploads/3.1 Method of Plant Disease Control_s35_img1.png]

Soil solarization

The soil surface is covered with transparent plastic film, taking advantage of solar radiation to heat the soil to temperatures that are lethal to many fungal pathogens and nematodes.

Soil steam sterilization

Soil in open fields or greenhouses is sterilized using steam. The steam kills plant pathogens (bacteria, fungi, viruses) by causing vital cellular proteins to denature.

Slide 36

Flooding

Temporary flooding is used as an effective method to control plant-parasitic nematodes and other soilborne pathogens.

Burning crop and pruning residues

For diseases for which no cure exists, infected crop and pruning residues are burned as a last resort option. Burning is an effective control of all kinds of pathogens

<div id="3-2-biological-control-of-plant-diseases"></div>

Lecture: 3.2 Biological Control of Plant Diseases

Course Slides: 3.2 Biological Control

Slide 1

Biological control of plant pathogens

Christine Roath

Slide 2

Overview

What is biological control, what are the benefits to its use

Mechanism of biological control

Requirements of successful biocontrol

Working example of biocontrol

Slide 3

What is biological control?

First coined by Harry Smith in relation to the biological control of insects

Suppression of insect populations by native or introduced enemies

Generic terms

A population-leveling process in which the population of one species lowers the number of another

Slide 4

Why use biological control?

WHEN :

Biological control agents are

Expensive

Labor intensive

Host specific

WHILE :

Chemical pesticides are:

cost-effective

easy to apply

Broad spectrum

Slide 5

Q

Why use biological control?

WILL:

Chemical pesticides

Implicated in ecological, environmental, and human health problems

Require yearly treatments

Broad spectrum

Toxic to both beneficial and pathogenic species

BUT:

Biological control agents

Non-toxic to human

Not a water contaminant concern

Once colonized may last for years

Host specific

Only effect one or few species

Slide 6

[Image not found: /uploads/3.2 Biological Control_s6_img1.png]

Q

Slide 7

Q

[Image not found: /uploads/3.2 Biological Control_s7_img1.png]

Slide 8

Q

Antagonism

Definition:

Antagonism refers to the inhibitory relationship where one organism suppresses or harms another, often by competition, production of antibiotics, or other mechanisms.

Example:

The bacterium *Bacillus subtilis* produces antibiotics that inhibit the growth of the fungal pathogen *Fusarium oxysporum*, which causes wilt in many crops.

Slide 9

Q

Antagonist

Definition:

An antagonist is the organism that suppresses or inhibits the growth or activity of a pest or pathogen through various mechanisms like competition, parasitism, or production of harmful substances.

Example:

Trichoderma harzianum is a fungal antagonist that suppresses the growth of plant-pathogenic fungi like *Rhizoctonia solani* by outcompeting them for nutrients and space or by directly attacking them.

Slide 10

Mechanisms of biological control of plant pathogens/ How do antagonists act?

Q

Antibiosis – Inhibition of one organism by another as a result of diffusion of an antibiotic

Antibiotic production common in soil-dwelling bacteria and fungi

Example: zwittermicin A production by *B. cereus* against *Phytophthora* root rot in alfalfa

Slide 11

Q

Mechanisms of biological control of plant pathogens

Nutrient competition – competition between microorganisms for carbon, nitrogen, O₂, iron, and other nutrients

Most common way organisms limit growth of others

Example

P. fluorescens, VITCUS, prevents bacterial blotch by competing with *P. tolaasii*

Slide 12

Q

Mechanisms of biological control of plant pathogens

Destructive mycoparasitism – the parasitism of one fungus by another

Direct contact

Cell wall degrading enzymes

Some produce antibiotics

Example

Trichoderma harzianum, BioTrek, used as seed treatment against pathogenic fungus

Slide 13

Q

[Image not found: /uploads/3.2 Biological Control_s13_img1.png]

[Image not found: /uploads/3.2 Biological Control_s13_img2.png]

Slide 14

Q

[Image not found: /uploads/3.2 Biological Control_s14_img1.png]

[Image not found: /uploads/3.2 Biological Control_s14_img2.png]

[Image not found: /uploads/3.2 Biological Control_s14_img3.png]

Slide 15

Requirements of successful biocontrol

Highly effective biocontrol strain must be obtained or produced

Be able to compete and persist

Be able to colonize and proliferate

Be non-pathogenic to host plant and environment

[Image not found: /uploads/3.2 Biological Control_s15_img1.wmf]

Slide 16

Requirements of successful biocontrol

Inexpensive production and formulation of agent must be developed

Production must result in biomass with excellent shelf life

To be successful as agricultural agent must be

Inexpensive

Able to produce in large quantities

Maintain viability

Slide 17

Requirements of successful biocontrol

Delivery and application must permit full expression of the agent

Must ensure agents will grow and achieve their purpose

[Image not found: /uploads/3.2 Biological Control_s17_img1.png]

Coiling of *Trichoderma* around a pathogen. (Plant Biocontrol by *Trichoderma* spp. Ilan Chet, Ada Viterbo and Yariv Brotman)

Slide 18

Plant pathogen control by *Trichoderma* spp.

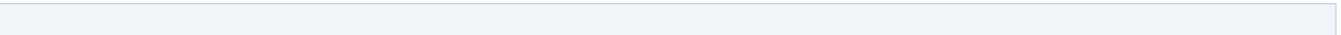
Trichoderma spp. are present in nearly all agricultural soils

Antifungal abilities have been known since 1930s

Mycoparasitism

Nutrient competition

Agriculturally used as biocontrol agent and as a plant growth promoter



[Image not found: /uploads/3.2 Biological Control_s18_img1.jpg]

<http://www.ars.usda.gov/is/pr/2002/021231.trichoderma.jpg>

Slide 19

Plant pathogen control by *Trichoderma* spp.

Why buy/develop a product that is readily available in the soil?

Slide 20

Plant pathogen control by *Trichoderma* spp.

Genetic Modification

Wild strains

Heterokaryotic – contain nuclei of dissimilar genotypes within a single organism

Biocontrol strains

Homokaryotic – contain nuclei which are similar or identical

Allows genetic distinction and non-variability

IMPORTANT FOR QUALITY CONTROL

Slide 21

Plant pathogen control by *Trichoderma* spp.

Most strains have innate resistance to some agricultural chemicals

Resistance is variable

Strains available for commercial use are selected or modified for resistance to specific chemicals

Slide 22

Plant pathogen control by *Trichoderma* spp.

How is it applied?

Favored by presence of high levels of plant roots

Some are highly rhizosphere competent

Capable of colonizing the expanding root surface

Can be used as soil or seed treatment

[Image not found: /uploads/3.2 Biological Control_s22_img1.jpg]

<http://www.nysaes.cornell.edu/ent/biocontrol/pathogens/images/trichoderma3.jpg>

Slide 23

Plant pathogen control by *Trichoderma* spp.

Action against pathogenic fungi

[Image not found: /uploads/3.2 Biological Control_s23_img1.jpg]

Attachment to the host hyphae by coiling

Lectin-carbohydrate interaction

(Hubbard et al., 1983. *Phytopathology* 73:655-659).

Slide 24

Plant pathogen control by *Trichoderma* spp.

Action against pathogenic fungi

[Image not found: /uploads/3.2 Biological Control_s24_img1.jpg]

2. Penetrate the host cell walls by secreting lytic enzymes

Chitinases

Proteases

Glucanases

(Ilan Chet, Hebrew University of Jerusalem).

Slide 25

Plant pathogen control by *Trichoderma* spp.

Some strains colonize the root with mycoparasitic properties

Penetrate the root tissue

Induce metabolic changes which induce resistance

Accumulation of antimicrobial compounds

Slide 26

Plant pathogen control by *Trichoderma* spp.

Commercial availability

T-22

Seed coating, seed pieces, transplant starter

Protects roots from diseases caused by *Pythium*, *Rhizoctonia* and *Fusarium*

Interacts with the Rhizosphere, near the root hairs and increases the available form of nutrients needed by plants.

Slide 27

Plant pathogen control by *Trichoderma* spp.

Future developments

Transgenes

Biocontrol microbes contain a large number of genes which allow biocontrol to occur

Cloned several genes from *Trichoderma* as transgenes

Produce crops which are resistant to plant diseases

Currently not commercially available

Slide 28

Questions

Slide 29

References

Current Microbiology Vol. 37 (1998), pp.6-11 Target Range of Zwittermicin A, and Aminopolyol antibiotic from *B. cereus*

Trichoderma for Biocontrol of Plant Pathogens: From Basic Research to Commercialized Products Gary E. Harman
Departments of Horticultural Science and of Plant Pathology ,Cornell University

Plant Biocontrol by Trichoderma spp. Ilan Chet, Ada Viterbo and Yariv Brotman. Department of Biological Chemistry

Trichoderma spp., including *T. harzianum*, *T. viride*, *T. koningii*, *T. hamatum* and other spp. by G. E. Harman, Cornell University, Geneva, NY 14456

The Plant Cell, Vol. 8, 1855-1869, October 1996 O 1996 American Society of Plant Physiologists Biocontrol of Soilborne Plant Pathogens. Jo Handelsman' and Eric V. Stabb

BioWorks products <http://www.bioworksbioccontrol.com/productsections/agprod.html>

Trichoderma image <http://www.ars.usda.gov/is/pr/2002/021231.trichoderma.jpg>

Trichoderma colonization image <http://www.nysaes.cornell.edu/ent/biocontrol/pathogens/images/trichoderma3.jpg>

www.weizmann.ac.il/Biological_Chemistry/scientist/Chet/Chet.html

<div id="3-3-chemical-control-fungicides"></div>

Lecture: 3.3 Chemical Control & Fungicides

Course Slides: 3.3 Chemical Control

Slide 1

Chemicals Used in Plants Disease Control

[Image not found: /uploads/3.3 Chemical Control_s1_img1.png]

Slide 2

Use of chemical in plant disease control

Designed to control plant diseases by inhibiting the growth of or by killing the disease-causing pathogens.

Prevent or reduce infections by utilizing various principles of disease control.

[Image not found: /uploads/3.3 Chemical Control_s2_img1.png]

Slide 3

History

[Image not found: /uploads/3.3 Chemical Control_s3_img1.png]

Slide 4

History

Prior to 1882: Disease control with element sulfur and copper

From 1882 to 1934: Disease control based on organo-metalics (fixed or organo-copper)

1934: Modern era of organic fungicides began with the dithiocarbamates

1943: Greatly improved fungicidal activity

Slide 5

Chemical control

Slide 6

Chemical used in plant disease control

Slide 7

Fungicides

Fungicide Mobility

Any toxic substance used to kill or inhibit the growth and reproduction of fungi is called fungicides.

Apoplastic movement- Xylem- mobile fungicides are absorbed by the plant and translocated upward to other tissues.

Symplastic Movement downward, with carbohydrates in the phloem.

Amphimobile fungicides move throughout the plant.

Translaminar fungicides can move across and through the leaves, but are not truly systemic, that travels only shorter distances in the plant.

Slide 8

What is apoplastic and symplastic transport mechanism?

The symplast pathway involves protoplasts, which are found within the living cytoplasm cells. Eg. Carbendazim

The apoplast pathway is a non-living route since the water solution moves in the spaces between cells and along the cell wall (no protoplasts involved). Eg. Benomyl (Bengimidazol)

Fungicide Mobility

[Image not found: /uploads/3.3 Chemical Control_s8_img1.png]

Slide 9

What is the difference between Translaminar and systemic?

Translaminar control agents penetrate into the plant tissue and are moved within a plant organ such as a leaf, but do not travel to other parts of the plant. For example, the chemical may move from the upper to lower surface of a leaf.

Systemic control agents can be effective at sites other than where they are sprayed.

Fungicide Mobility

[Image not found: /uploads/3.3 Chemical Control_s9_img1.png]

Slide 10

1. Apoplastic Movement (acropetal):

Move upward from the point of entry through the plant's xylem. e.g. Tebuconazole, Propamocarb, Benomyl.

4. Translaminar fungicides (locally systemic): Absorbed by leaves and move through the leaf to the opposite surface. e.g. Dimethomorph, Iprodione, etc.

2. Symplastic Movement (basipetal):

Move downward, with carbohydrates in the phloem. e.g. Trifloxystrobin, Carbendazim.

3. Amphimobile fungicides (truly systemic):

Move both upward through the xylem, and downward through the phloem. e.g. Fosetyl Al, Phosphoric acid.

Slide 11

Inhibit fungal growth

kill fungal pathogens

Prevent damage caused by fungi

Inhibit spore production

Slide 12

Fungicides

Chemical nature

Mode of action

General use

Slide 13

Classification of fungicides based on chemical nature

Sulphur : zineb, maneb

Copper: bordeaux mixture

Mercurial: phenyl mercury acetate

Quinone: dichlone

Heterocyclic nitrogenous: captan

Oxathiins: carboxin

Benzimidazoles: benomyl MBC

Fungicides in plant disease control, Y.L. Nene and P.N. Thapliyal (third edition).

Slide 14

[Image not found: /uploads/3.3 Chemical Control_s14_img1.jpg]

Slide 15

Classification of fungicides based on general use

Seed protectant

Soil fungicides

Foliage and blossom

Fruit protectant

Slide 16

[Image not found: /uploads/3.3 Chemical Control_s16_img1.jpg]

Q

Slide 17

Q

Classification of fungicides based on mode of action

[Image not found: /uploads/3.3 Chemical Control_s17_img1.png]

Ex: Thiram

Ex: Tilt

Ex: Bordeaux Mixture

Slide 18

[Image not found: /uploads/3.3 Chemical Control_s18_img1.png]

[Image not found: /uploads/3.3 Chemical Control_s18_img2.png]

Slide 19

Q

[Image not found: /uploads/3.3 Chemical Control_s19_img1.png]

Slide 20

[Image not found: /uploads/3.3 Chemical Control_s20_img1.png]

Systemic

[Image not found: /uploads/3.3 Chemical Control_s20_img2.png]

Slide 21

[Image not found: /uploads/3.3 Chemical Control_s21_img1.png]

[Image not found: /uploads/3.3 Chemical Control_s21_img2.png]

Propiconazole

(1ml/lit) may be applied. Spray

Carbendazim 250 g or

Chlorothalonil 1 kg or

Edifenphos 1 lit/ha

[Image not found: /uploads/3.3 Chemical Control_s21_img3.png]

Systemic

Rice sheath blight

[Image not found: /uploads/3.3 Chemical Control_s21_img4.png]

[Image not found: /uploads/3.3 Chemical Control_s21_img5.png]

iprodione, propiconazole, azoxystrobin, and carbendazim are effective in management of brown spot disease (Mandal and Jha, 2008)

Brown spot of rice

Slide 22

[Image not found: /uploads/3.3 Chemical Control_s22_img1.png]

[Image not found: /uploads/3.3 Chemical Control_s22_img2.png]

Slide 23

[Image not found: /uploads/3.3 Chemical Control_s23_img1.png]

Slide 24

[Image not found: /uploads/3.3 Chemical Control_s24_img1.png]

Vitavax-200 or Provex-200, Sulfex, Thiovit, Mancozeb (Diathane M-45)

Non-systemic

[Image not found: /uploads/3.3 Chemical Control_s24_img2.png]

Blitox, Fytolan

Slide 25

[Image not found: /uploads/3.3 Chemical Control_s25_img1.jpg]

Slide 26

Bactericides

Slide 27

Bactericides

[Image not found: /uploads/3.3 Chemical Control_s27_img1.png]

[Image not found: /uploads/3.3 Chemical Control_s27_img2.png]

[Image not found: /uploads/3.3 Chemical Control_s27_img3.png]

Soil borne bacterial diseases like bacterial wilt, black rot, Bacterial Leaf spot etc.

Controlling bacterial blight

controlling rice leaf blight caused by *Xanthomonas oryzae* PV. *oryzae*

Slide 28

Nematicides used in Bangladesh

Nematicide	Trade Name	Disease
Diazinon	Basudin 10 G	Ufra disease of rice
Carbofuran	Carbofuran 3G	White tip of rice

Fipronil	Fipronil 0.3 G	White tip of rice
Carbosulfan	Benefit 20 EC	Ufra disease of rice
Carbofuran	Furadan 5 G	Root knot disease
Carbofuran	Furadan 5 G	Wheat gall nematode

(Bangladesh crop protection association, 2021)

Slide 29

Q

Insects Involved in Viral Disease

Aphid

Sap-sucking insects

BCMV, BYMV, CMV

Cypermethrin, deltamethrin, dimethoate.

[Image not found: /uploads/3.3 Chemical Control_s29_img1.jpg]

Whitefly

Piercing sucking mouthparts

TYLCV, MYMV

Imidacloprid, thiamethoxam, Sobicron

Other Control Measures:

Yellow sticky trap.

Biological control (Ladybug).

Malathion (0.2% solution).

[Image not found: /uploads/3.3 Chemical Control_s29_img2.jpg]

Slide 30

CHEMICAL

NEMATOCIDES

FUNGICIDES

RICE ROOT-KNOT BY NEMATODE

NEGATIVE EFFECTS OF FUNGICIDES & NEMATOCIDES

LATE BLIGHT OF POTATO

BLAST OF RICE

Caused by the fungus-like pathogen *Phytophthora infestans*.

CAUSED BY *MAGNAPORTHE ORYZAE*

CAUSED BY

MELOIDOGYNE spp.

1 .ENVIRONMENTAL POLLUTION

2 .DEVELOPMENT OF PATHOGEN RESISTANCE

3 .RESIDUAL TOXICITY IN CROPS

4. NON TARGET ORGANISM HARM

5 .HUMAN AND ANIMAL HEALTH RISK

COMMON NEMATOCIDES:

COMMON FUNGICIDES:

1. Mancozeb + Metalaxyl (Ridomil Gold)

2. Copper Oxychloride (Blitox)

COMMON FUNGICIDES:

1. TRICYCLAZOLE

2 PROPICONAZOLE

3, ISOPROTHIOLANE (FUNGICIDE).

1.CARBOFURAN (FURADAN)

2.FOSTHIAZATE

3.OXAMYL (VYDATE)

Slide 31

Formulation

[Image not found: /uploads/3.3 Chemical Control_s31_img1.jpg]

Water dispersible powder

(Agrizeb, Mancovit, Ipromil etc.)

[Image not found: /uploads/3.3 Chemical Control_s31_img2.jpg]

Dust

(Copper, Sulphur)

[Image not found: /uploads/3.3 Chemical Control_s31_img3.jpg]

Suspension

(Cuproxat, Azonil, Medallion etc.)

Slide 32

Formulation

[Image not found: /uploads/3.3 Chemical Control_s32_img1.png]

Emulsifiable concentrate

(Tilth, Aconazol, Treat etc.)

[Image not found: /uploads/3.3 Chemical Control_s32_img2.jpg]

Wettable powder

(Captan, Goldazim, Mancer etc.)

[Image not found: /uploads/3.3 Chemical Control_s32_img3.jpg]

Flowable

(Phytovit, Agrovit etc.)

Slide 33

Methods of Application

Seed treatment

[Image not found: /uploads/3.3 Chemical Control_s33_img1.png]

Steep treatment

(Agrizeb, mancovit)

[Image not found: /uploads/3.3 Chemical Control_s33_img2.jpg]

Sprinkle treatment

(Ipromil, Azonil)

Slide 34

Seed treatment

[Image not found: /uploads/3.3 Chemical Control_s34_img1.jpg]

Dust treatment

(sulphur, copper)

[Image not found: /uploads/3.3 Chemical Control_s34_img2.jpg]

Slurry treatment

(Captan, mancer)

Slide 35

Seed treatment

[Image not found: /uploads/3.3 Chemical Control_s35_img1.jpg]

Fumigation

(methyl bromide, chloropicrin)

[Image not found: /uploads/3.3 Chemical Control_s35_img2.jpg]

Pelleting

(Tween, carbendazim)

Slide 36

Methods of Application

Soil treatment

[Image not found: /uploads/3.3 Chemical Control_s36_img1.jpg]

[Image not found: /uploads/3.3 Chemical Control_s36_img2.jpg]

Furrow application

Soil drenching

[Image not found: /uploads/3.3 Chemical Control_s36_img3.jpg]

Fumigation

Slide 37

Methods of Application

General precautions and suggestions concerning fungicidal soil treatment

Soil conditions

Soil amendments

Treating tools

Time interval between treatment and planting

Slide 38

Methods of Application

[Image not found: /uploads/3.3 Chemical Control_s38_img1.jpg]

[Image not found: /uploads/3.3 Chemical Control_s38_img2.jpg]

Dusting

Spraying

Slide 39

Q

Characteristics of a Good Fungicide

High field performance.

Low phytotoxicity.

Low toxicity to human beings.

Storage stability.

Stability after dilution.

Available in different formulations.

Cheaper.

Easy to transport.

Slide 40

Factors Affecting to Fungicide Application

Climatic condition

Air current

Crop phenology

Cost/benefit ratio

Revision Questions & Detailed Answers

Introduction & Pathogenicity

Q1: [♦ Repeated ×4] What is parasitism? What is pathogenicity and parasitism?

■ Source: Original Course Content

Answer

- Parasite: An organism that lives on and derives food from another organism.
- Parasitism: The relationship between a parasite and its host is called parasitism.
- Pathogen: An agent that can incite a disease.
- Pathogenicity: It is the ability of the pathogen to cause disease.

Q2: [♦ Repeated ×3] "Symbiotic relationship is a parasitism" — explain with example

■ Source: Original Course Content

Answer

- Symbiotic relationship refers to any close and long-term interaction between two organisms, including mutualism, commensalism, and parasitism.
- Parasitism is a specific type of symbiosis where one organism, the parasite, benefits by exploiting the host for nutrients, water, or shelter, causing harm to the host.

Pathogen like fungi (*Puccinia graminis*) is a plant parasite. These pathogens attach to, penetrate, infect, and invade host tissues, leading to symptoms such as necrosis, wilting, and reduced yield.

While all parasitism is symbiotic, not all symbiosis is parasitic — mutualism and commensalism do not harm the host. For example, mycorrhizae and root nodule bacteria show mutualism — they are symbiotic but not parasitic.

Q3: "All pathogens are parasites but all parasites are not pathogens" — explain

■ Source: Original Course Content

Answer

A parasite is not necessarily a pathogen, and a pathogen is not necessarily a parasite. The relationship between parasitism and pathogenicity is not absolute.

1. Parasites without Pathogenicity

Some organisms are biological parasites but do not cause disease. For example, **mycorrhizae** or **root nodule bacteria** derive nutrients from the host plant but do not harm it, instead engaging in a highly beneficial mutualistic relationship.

2. Pathogens without Parasitism

Conversely, some pathogens can cause severe plant disease without being true active parasites. They may act from the outside through lethal secreted **toxins** or cell-wall degrading enzymes without living on or inside the host tissue.

Conclusion:

Therefore, while all pathogens interact with the host in a parasitic manner to extract nutrients, the clinical manifestation of disease (pathogenicity) is not a requirement for all parasitic relationships.

Q4: "Most but not all pathogens are parasites, most but not all parasites are pathogens" — explain

■ Source: Original Course Content

Answer

- Parasite: An organism that lives on and derives food from another organism.
- Pathogen: An agent that can incite a disease.

Most pathogens are parasites because they live in/on the host and derive nutrition while causing disease (e.g., *Puccinia graminis* causing rust). However, not all pathogens are parasites — some cause disease through toxins without direct parasitic relationship.

Similarly, most parasites cause some harm (making them pathogens), but some parasites like mycorrhizal fungi and *Rhizobium* in root nodules are beneficial — they are parasites but not pathogens.

Q5: Write a short note on necrotrophic parasite

■ Source: Original Course Content

Answer***Necrotrophs (Facultative parasites) — Biochemical and morphological features:***

- Host cell rapidly killed
- Toxins and cytolytic enzymes produced
- No special parasitic structures formed
- Host penetration via wounds or natural openings

Ecological features:

- Wide host range
- Able to grow saprophytically away from the host
- Attack juvenile, debilitated or senescing tissues

Q6: Compare between biotrophs and necrotrophs

■ Source: Original Course Content

Answer***Comparison: Biotrophs vs. Necrotrophs***

Feature	Biotrophs	Necrotrophs
Host cell killing	Not rapidly killed	Rapidly killed
Toxins & enzymes	Few or none produced	Toxins and cytolytic enzymes produced
Special structures	Haustoria formed	No special structures
Host range	Narrow	Wide
Growth away from host	Unable	Able (saprophytic)
Penetration	Direct or natural openings	Wounds or natural openings
Host attacked	Healthy hosts at all stages	Juvenile, debilitated or senescing tissues

Pathogenesis & Direct Penetration

Q1: [♦ Repeated ×2] What is meant by pathogenesis? Diagrammatically represent the disease cycle

■ Source: Original Course Content

Answer

Pathogenesis: The chain of events involved in disease development, including the stages of development of the pathogen and the effect of disease on the host is called disease cycle or pathogenesis.

Stages in the development of disease:

- Inoculation
- Pre-penetration Phenomena
- Penetration
- Infection
- Invasion
- Growth and Reproduction of the Pathogen
- Dissemination of the Pathogen
- Over-wintering / over-summering of the Pathogen

Diagram (Sequential Cycle):

- Inoculum → Pre-penetration → Penetration → Infection → Invasion → Reproduction → Dissemination → Overwintering → [back to Inoculum]

Q2: What is meant by disease cycle, parasitism and obligate parasite?

■ Source: Original Course Content

Answer

- Disease cycle: The chain of events involved in disease development is called a disease cycle or pathogenesis.
- Parasitism: The relationship between a parasite and its host is parasitism.

Obligate parasites (Biotrophs): Pathogens that can only live on a living host and cannot grow away from it. They have a narrow host range, attack healthy hosts at all stages of development, form special parasitic structures such as haustoria, and produce few or no toxins.

Q3: What do you mean by pathogenesis? If penetration phase does not occur in disease cycle then what will happen?

■ Source: Original Course Content

Answer

Pathogenesis: The chain of events involved in disease development is called pathogenesis or disease cycle.

If penetration does not occur:

- No establishment of pathogen in host tissue
- Disruption of disease cycle
- Limited pathogen spread
- No infection or disease development
- Pathogen survival is jeopardized
- The host remains unaffected and the disease cycle is effectively interrupted.

Q4: Which are inocula? Why are all propagules not inocula?

■ Source: Original Course Content

Answer

Inoculum: The pathogen or its parts that can cause infection; that portion brought into contact with the host.

Inocula include:

- Fungi: spores/conidia, mycelium, sclerotia
- Bacteria and viruses: individual cells/particles
- Nematodes: the organism itself or its eggs
- Parasitic flowering plants: seeds
- Why all propagules are NOT inocula: Not all propagules land on a susceptible host or reach a suitable environment. When displaced pathogen does not get any medium for survival (e.g., falls on a stone), it is called ineffective dispersal — so it is a propagule but not an effective inoculum. Only when displacement occurs on a susceptible host under suitable environmental conditions does the propagule act as true inoculum (effective dispersal).

Q5: Differentiate between infection and invasion with examples

■ Source: Original Course Content

Answer**Comparison: Infection vs. Invasion**

Feature	Infection	Invasion
Definition	Establishment of contact by the pathogen with the host plant.	Spread of the pathogen into the host after infection.
Process	Pathogen grows/multiplies, colonizes host.	Pathogen moves through host tissues.
Result	Appearance of symptoms (or latent infection).	Systemic or local spread.
Example	<i>Phytophthora infestans</i> infects potato	<i>Apple scab</i> (<i>Venturia inaequalis</i>) grows between cuticle and epidermis.

Q6: Illustrate the direct penetration process of fungus into host cell

■ Source: Original Course Content

Answer**Direct penetration through intact plant surfaces (fungi, nematodes, parasitic plants):**

- The penetration is through mechanical force and softening of the cell walls by an enzyme.
- Most fungi form an appressorium at the end of the germ tube.
- A penetration peg emerges from the flat surface of the appressorium and pierces the cuticle and cell wall.
- The peg grows into fine hyphae then reaches normal diameter once inside the cell.

Q7: [♦ Repeated ×2] Explain the mechanism of using physical weapons by fungi and nematodes to penetrate directly into the host surface

■ Source: Original Course Content

Answer

Fungi: Form an appressorium at the end of the germ tube. A penetration peg emerges from the flat surface and pierces the cuticle and cell wall using both mechanical force and cell-wall-softening enzymes. The peg grows into fine hyphae inside the cell.

Nematodes: Penetrate using their stylet (a needle-like structure). Insert the stylet or the entire nematode enters the plant tissue. They are attracted to the root by CO₂ and some amino acids associated with root growth.

Toxins in Disease Development

Q1: Distinguish among pathotoxin, vivotoxin and phytotoxin

■ Source: Original Course Content

Answer

Comparison: Vivotoxin, Pathotoxin, and Phytotoxin

Feature	Vivotoxin	Pathotoxin	Phytotoxin
Definition	Substance produced in infected host by pathogen/host that functions in disease but is not the initial inciting agent.	Chemical of biological origin; host-specific toxin producing almost all disease symptoms even in the absence of the causal organism.	Poisonous substances synthesized by plants or plant pathogens through biochemical reactions.
Host Specificity	Non-specific	Host specific	Defensive agent / Non-specific
Role in Disease	Partial role	Causal/primary role	Defensive/protective role
Examples	Fusaric acid, Pyricularin	T-toxin, Victorin, HC-toxin, Wildfire toxin	Against pests, insects, environmental stress

Q2: [♦ Repeated ×3] Note down the target site of toxins in plant cells / Role of toxins in disease development

■ Source: Original Course Content

Answer

Toxins are injurious substances secreted by micro-organism host complex that act on living host protoplast to affect disease symptoms. Mechanisms of phytotoxicity:

1. Changes in Cell Wall Permeability:

- Toxins alter permeability of plasma membrane
- Cause loss of water and electrolyte leakage (K^+ , Na^+ , Ca^{2+} , Cl^- , PO_4^{3-} , Mg^{2+})
- Allow unrestricted entry of substances including the toxin itself

2. Disruption of Normal Metabolic Processes:

- Toxins affect the cellular transport system of H^+ / K^+ exchange in cell membrane
- Disrupt potassium-regulated enzyme activity and photosynthesis/ATP production

3. Malfunctioning of Enzyme System:

- Tabtoxin: inactivates glutamine synthetase → ammonia accumulates → destroys thylakoid membranes → chlorosis and necrosis
- Phaseolotoxin: inactivates ornithine carbamoyl transferase → accumulates ornithine, depletes arginine
- Tentoxin: inhibits polyphenol oxidases

4. Interference with Growth Regulatory System:

Cause hormonal imbalance → overgrowths, rosetting, excessive root branching, leaf epinasty, defoliation

5. Inhibit Root Growth:

Fusarium moniliforme produces a thermostable toxin causing browning of roots and restricts development

Q3: [♦ Repeated ×2] Write down the toxin hypothesis in disease development / State the toxin hypothesis with an example

■ Source: Original Course Content

Answer

Toxin Hypothesis (Vivotoxin concept — Diamond & Waggoner, 1953):

- A vivotoxin is defined as a substance produced in the infected host by the pathogen and/or its host, which functions in the production of disease but is not itself the initial inciting agent of disease.

Criteria:

- Reproducible separation of the toxin from the sick plant
- Purification or chemical characterization of the toxin
- Induction of at least a portion of the disease symptoms by placing the toxin in a healthy plant
- Example: Fusaric acid and pyricularin are typical vivotoxins — produced inside the infected host and contribute to disease symptoms, but they are not the initial cause of infection.

Q4: State the criteria of vivotoxin

■ Source: Original Course Content

Answer

Criteria of vivotoxin (same as toxin hypothesis criteria):

- Reproducible separation of the toxin from the sick plant
- Purification or chemical characterization
- Induction of at least a portion of the disease symptoms by placing the toxin in a healthy plant
- Typical examples: Fusaric acid and pyricularin

Pathophysiology

Q1: [♦ Repeated ×2] "Pathogens interfere with the normal mechanism of photosynthesis of a plant" — explain

■ Source: Original Course Content

Answer

- Damage to leaf tissues — reduced photosynthetic area; e.g., blights and leaf spots
- Destruction of chloroplasts — some diseases destroy chloroplasts and cause their disintegration
- Toxin production — pathogens produce toxins such as tentoxin and tabtoxin which inhibit enzymes involved in photosynthesis
- Vascular pathogens — reduce chlorophyll content; photosynthesis stops before wilting occurs
- Viruses and mollicutes — cause chlorosis and stunting; photosynthesis is greatly reduced

Q2: How does pathogen interfere with the translocation of organic nutrient through phloem?

■ Source: Original Course Content

Answer

- Pathogens may interfere with movement of nutrients from leaf cells to phloem.
- Rusts and mildews cause accumulation of photosynthetic products in areas invaded by pathogens.
- Infected areas are characterized by reduced photosynthesis and increased respiration.
- Canker causes pathogens to attack the stem and remain confined there.
- Pathogen may attack and destroy phloem elements — interfering with downward translocation of nutrients.

Q3: How does nutrient imbalance influence plant predisposition?

■ Source: Original Course Content

Answer

- Disease caused by biotrophs are favored by good nutrient status but disease caused by necrotrophs are favored by poor soil status.
- Higher nitrogenous fertilizer application leads to dense canopy, predisposing plants to disease.
- Balanced fertilizer application is treated as a control measure to kill or inhibit pathogens in the field.
- Vascular pathogens affect movement of nutrients from soil to leaves — rest of plant is deprived of nutrients, making it predisposed to further disease.

Dissemination of Pathogens

Q1: [♦ Repeated ×2] "Dissemination is an important phenomenon of disease spread" — explain / Importance and ways of plant pathogen dissemination

■ Source: Original Course Content

Answer

Definition: Displacement of a plant pathogen from its place of production/origin to a suitable place where it can grow/establish; transport of spores or infectious bodies from one host to another resulting in spread of disease.

Importance of Dissemination:

- Necessary for food and nutrition of pathogen
- Ensures survival of pathogen
- Allows pathogen to complete its life cycle
- Enables reproduction and continuation of the species

Ways/Agents of Dissemination:

- Wind, Water, Soil, Seed
- Animal, Human, Machinery, Transport system
- Insects and other vectors
- Phanerogamic parasites (e.g., Dodder/Cuscuta)

Q2: Describe the process of plant disease dissemination by air

■ Source: Original Course Content

Answer***Characteristics of wind-disseminated fungi:***

- Numerous spores, discharged with sufficient force
- Light in weight, smaller size, thick outer wall, less affinity to water

Short distance dissemination:

- Sporangia of downy mildew fungi, conidia of powdery mildew fungi, and basidiospores of rust fungi

Long distance dissemination:

- Uredospores of rust fungi, Chlamydospores of smut fungi and conidia of Alternaria, Helminthosporium and Pyricularia

Examples:

- P. graminis tritici — Horizontal: 600 km, Altitude: 500 m above infected field
- P. infestans — 200 m to 600 km horizontal
- Mold spores (Aspergillus, Penicillium, Cladosporium) — altitude up to 2000 m

Factors affecting wind dissemination:

- Periodicity (diurnal/nocturnal)
- Air current (steady-horizontal, breeze and turbulence-vertical, irregular-multidimensional)

Epidemiology & Disease Triangle

Q1: [♦ Repeated ×2] Define epidemic and endemic / Define endemic, pandemic and sporadic. Write down the elements of epidemic

■ Source: Original Course Content

Answer

Epidemic (Epiphytotic): A sudden outbreak of disease periodically over a widespread area in devastatingly severe form causing severe losses. E.g., wheat stem rust (*Puccinia graminis tritici*), late blight of potato — Irish famine 1845–1846, rice blast — Bengal famine 1943.

Endemic (Enphytotic): When a disease is more or less constantly occurring year after year in a moderate to severe form in a locality. E.g., wart disease of potato (*Synchytrium endobioticum*) in Darjeeling; citrus canker in Asia.

- **Pandemic:** When an epidemic spreads over continents/subcontinents and involves mass mortality. E.g., black stem rust of wheat in India during 1947.
- **Sporadic:** Diseases occurring at irregular intervals over limited areas. E.g., Fusarium wilt of cotton, loose smut of wheat.

Elements of Epidemic (Disease Triangle + Time + Humans):

- Susceptible host
- Virulent pathogen
- Favorable environment
- Time
- Human influence — These interactions are described by the disease pyramid

Q2: In epidemiological study disease development differs by monocyclic and polycyclic model — explain with example

■ Source: Original Course Content

Answer***Monocyclic (simple interest model):***

- Pathogens complete one disease cycle per year
- Primary inoculum is the only inoculum for the entire season — no secondary inoculum/secondary infection
- Disease progress represented as saturation curve
- Equation: $X = Q \cdot R \cdot t$ (Vanderplank)
- Examples: Loose smut of wheat, Blackleg of potato (*Erwinia caratovora*), Verticillium wilt

Polycyclic (compound interest model):

- Pathogens go through 2–30 disease cycles in one growing season
- Large amounts of asexual spores produced as secondary inoculum at each infection site
- Each cycle multiplies inoculum manyfold
- Disease progress represented as sigmoid curve
- Equation: $X = X_0 \cdot e^{rt}$
- Examples: Late blight of potato, downy mildews, powdery mildews, grain rusts, aphid-borne viruses

Q3: Analyze the equation $X = X_0 e^{rt}$ of polycyclic disease

■ Source: Original Course Content

Answer

This is the Vanderplank polycyclic model (compound interest disease model):

- X = amount of disease at time t
- X_0 = initial amount of disease (initial inoculum)
- r = apparent infection rate (rate of disease increase)
- t = time (duration of epidemic)
- e = base of natural logarithm

Implications:

- If r is very high, reducing X_0 only delays the epidemic
- If r is very high, X_0 must be reduced to very low levels to have significant effect
- Reducing r has relatively greater effect on epidemic than reducing X_0
- Reducing X_0 makes strategic sense only if r is also low or being reduced
- Three ways to reduce disease: (1) Reduce X_0 , (2) Reduce r , (3) Reduce duration t

Q4: What is progressive epidemic? Illustrate disease gradient curve

■ Source: Original Course Content

Answer

Progressive epidemic: If the epidemic is considered as an increase of disease in extensity (spreading outward), that is called a progressive epidemic.

Disease Gradient Curve (DGC): The pattern of epidemic (in terms of number of lesions, amount of diseased tissues or number of diseased plants) as it spreads over distance is given by a curve called disease gradient curve.

What DGC reveals:

- The amount of disease is generally greater near the source of inoculum and decreases with increasing distance.
- The number of diseased plants and severity of disease decrease steeply until they reach zero.
- As distance from inoculum source increases, disease severity decreases.
- % Disease
- 100 | \.

50 | \.

- 0 | _____ \.
- Source Distance (m/km)

Q5: Differentiate among disease triangle, disease tetrahedron and disease pyramid with neat labeled diagram

■ Source: Original Course Content

Answer

Comparison: Disease Triangle, Disease Tetrahedron, and Disease Pyramid

Concept	Components / Dimensions	Core Description & Meaning
Disease Triangle	3 Components: 1. Susceptible Host 2. Virulent Pathogen 3. Favorable Environment	Represents the interaction of three essential factors. If any of these three components is zero or absent, no disease can occur. The area of the triangle represents the amount/severity of disease.
Disease Tetrahedron	4 Components: 1. Host 2. Pathogen 3. Environment 4. Time	Adds the fourth dimension of Time . Time represents when and for how long the favorable conditions exist, which is critical for epidemic progression (e.g., incubation period, rate of reproduction).
Disease Pyramid	5 Components: 1. Host 2. Pathogen 3. Environment 4. Time 5. Human Influence	Adds the fifth component of Human Influence . Humans influence the severity of plant disease through cultural practices, breeding resistant varieties, fungicide application, global transport of pathogens, etc.

Diagram Guidelines for Drawing:

- **Disease Triangle:** Draw a flat triangle. Label the three corners/sides as **Host (susceptible)**, **Pathogen (virulent)**, and **Environment (favorable)**. Write **Disease** in the center.
- **Disease Tetrahedron:** Draw a three-sided pyramid (tetrahedron) with a triangular base. Label the four vertices as **Host**, **Pathogen**, **Environment**, and **Time**.
- **Disease Pyramid:** Draw a four-sided pyramid (pentahedron) with a square base. Label the five vertices as **Host**, **Pathogen**, **Environment**, **Time**, and **Human**.

Q6: [♦ Repeated ×2] "Plant changes their susceptibility to disease with age" — explain with diagram

■ Source: Original Course Content

Answer

- Pattern 1a: Susceptible only during earliest growth period, become resistant during adult stage. Example: Pythium damping-off, downy mildew, systemic smuts.
- Pattern 1b: Susceptible only in the stages of maximum growth. Example: Rusts and viral infections.

Pattern II: Blossoms or fruits are resistant during growth and early adult period but become susceptible near ripening. Example: Infections due to Botrytis, Penicillium, Monilia, and Glomerella.

Pattern III: A juvenile susceptibility is followed by a period of relative resistance in early adult stage and then further susceptibility after maturity. Example: Potato late blight and tomato early blight.

Q7: Why moisture is an important factor for development of plant disease in epidemic form?

■ Source: Original Course Content

Answer

Moisture is the predominant factor in the development of most epidemics caused by fungi (blights, downy mildews, leaf spots, rusts, anthracnoses), bacteria (leaf spots, blights, soft rots), and nematodes.

Major Effects of Moisture:

- **Effects on Host:** Moisture promotes succulent, lush growth of the host, making it physically more susceptible to pathogen penetration.
- **Effects on Pathogen:**
 - Increases sporulation of fungi and rapid multiplication of bacteria.
 - Favors spore release by many fungi and oozing of bacterial cells.
 - Enables fungal spores to germinate and allows motile zoospores, bacteria, and nematodes to move through water films to reach hosts.
 - High soil moisture directly increases the activity and survival of soil-borne pathogens like *Pythium*, *Phytophthora*, *Rhizoctonia*, and *Plasmodiophora brassicae*.
- **Effects on Disease Cycle:** Different moisture levels influence different events — e.g., *Pyricularia oryzae* sporulates at moderate moisture (85–90%) but release of spores, dissemination, and infections require high moisture (92–100%).
- **Effects on Vector:** High humidity reduces the activity of insect vectors but enhances the activity of fungal and nematode vectors.

Q8: [♦ Repeated x2] Write down the Vanderplank's Equivalence theorem correlated with disease triangle

■ Source: Original Course Content

Answer

Vanderplank's Equivalence Theorem: "Effects of host, pathogen and environment can be translated into terms of the rate parameter of an epidemic."

Changes in any component has an equivalent effect on disease:

- More/less susceptible host
- More/less favorable environment
- More/less amount of aggressive pathogen = same proportional change in disease
- Correlation with Disease Triangle: The disease triangle has three sides: Host, Pathogen, and Environment. According to Vanderplank's theorem, any change in any side has an equivalent and proportional effect on the area of the triangle — which represents the amount of disease. Therefore, disease management principles are centered around manipulating one or more components of the disease triangle.

Q9: How do dense planting and excessive nitrogen application modify epidemiological factors?

■ Source: Original Course Content

Answer

Dense planting: Creates a humid, protected canopy environment that favors diseases that spread in moist conditions. It increases foliar density, which favors development of *Phytophthora infestans* during periods of favorable weather.

Excessive nitrogen application: Promotes succulent, lush plant growth — making the host more susceptible. Disease caused by biotrophs are favored by good nutrient status. Higher nitrogen leads to faster vegetative growth, providing more susceptible tissue for pathogen establishment and increasing the rate of epidemic (r).

Q10: How does the fluctuation of day and night temperature affect epidemic development?

■ Source: Original Course Content

Answer

Temperature directly dictates the geographic distribution and seasonal occurrence of plant disease epidemics.

Key Temperature Fluctuation Effects:**• Effects of Temperature on Hosts:**

- Temperatures higher or lower than the optimum range for the host plant reduce the plant's metabolic activity, making it stressed and predisposed to disease infection.

• Effects on Pathogen:

- Favorable temperature helps polycyclic pathogens complete their lifecycle within a very short time, exponentially increasing inoculum production.
- Low temperature reduces the viability of fungi and bacteria.
- High temperature reduces the viability of viruses and mycoplasmas.
- *Pyricularia oryzae* produces spores more rapidly at 28°C; in *Phytophthora infestans*, zoospore germination occurs at 10–12°C while direct conidial germination occurs at 24°C.

• Effects of Day/Night Temperature Cycles:

- Sporulation of *Pyricularia oryzae* occurs primarily during the daytime and decreases with the reduction of light.
- Dissemination of spores begins at night and ceases at sunrise.
- This day/night temperature and light cycle creates periodic peaks of inoculum that drive epidemic cycles.

Principles of Disease Control

Q1: [♦ Repeated ×2] Describe briefly the principles of plant disease control

■ Source: Original Course Content

Answer***The 6 principles (APTREE):***

1. Avoidance: Avoiding disease by altering planting time or planting in areas where inoculum is ineffective due to environmental conditions. Done through choice of geographical area, time of sowing, field selection, disease-escaping varieties.
2. Protection: Preventing infection by creating a chemical toxic barrier between plants and pathogens. Done through chemical treatment, control of insect vectors, modification of environment.
3. Therapy: Reducing the severity of disease in an already infected individual by chemicals (chemotherapy), tree surgery, heat therapy. E.g., Bordeaux Mixture.
4. Resistance (Resistant Varieties): Preventing infection or reducing effects by managing the host through genetic improvement. Includes selection, hybridization, and mutation for disease resistance.
5. Eradication: Eliminating a pathogen after it is introduced into an area but before it has widely spread. Done through crop rotation, soil treatment, burning infected residues, biological control.
6. Exclusion: Preventing inoculum from entering or establishing in the field. Done through seed treatment, inspection, certification, quarantine, cleaned farm equipment.

Q2: "Crop rotation is considered as ancient preventive measure of plant disease management" — explain with examples

■ Source: Original Course Content

Answer

Crop rotation is the most widely employed control measure in agriculture. It manipulates the environment to reduce pathogen inoculum levels by rotating crops and varieties over seasons.

Mechanism:

- When the same crop is grown year after year, soil-borne pathogens and their inoculum build up
- Rotation to a non-host crop deprives the pathogen of its food source
- Without a susceptible host, the pathogen population declines

Examples:

- Rotate soybean with corn to control soybean cyst nematode
- Fusarium wilt pathogens decline when susceptible crops are rotated with non-host crops
- Blackleg of potato (*Erwinia caratovora*) managed by not planting potato consecutively in same field

Q3: Mention the advantages of biological control over chemical control of plant disease

■ Source: Original Course Content

Answer***Advantages of Biological Control:***

- Non-toxic to human; not a water contaminant concern
- Once colonized may last for years (unlike chemicals requiring yearly treatments)
- Host specific — only affects one or few species, not broad spectrum
- Biocontrol agents give protection throughout the crop period
- Leaves no residual problem; application is safer to environment and to the applicator
- Can eliminate pathogens from the site of infection
- Not only controls disease but also enhances root and plant growth by encouraging beneficial soil microflora
- Increases crop yield
- Chemical pesticides are broad spectrum — toxic to both beneficial and pathogenic species — and implicated in ecological, environmental, and human health problems.

Q4: Differentiate between systemic and contact fungicides

■ Source: Original Course Content

Answer**Comparison: Systemic vs. Contact (Non-systemic) Fungicides**

Feature	Systemic Fungicides	Contact (Non-systemic) Fungicides
Movement	Moves inside plant through vascular bundles (xylem/phloem).	Remains on the plant surface where applied.
Mode	Effective at sites other than where sprayed (translocation).	Only effective where directly applied.
Action	Curative and protective.	Mainly protective/preventative.
Mobility Types	Apoplastic, Symplastic, Amphimobile, Translaminar.	No translocation.
Examples	Benomyl, Propiconazole, Carbendazim, Tricyclazole.	Bordeaux mixture, Mancozeb, Captan, Zineb, Thiram.

Q5: Enumerate the criteria of a good fungicide used in agriculture

■ Source: Original Course Content

Answer

- High field performance
- Low phytotoxicity
- Low toxicity to human beings
- Storage stability
- Stability after dilution
- Available in different formulations
- Cheaper
- Easy to transport

Q6: [♦ Repeated ×3] "Biological control is prioritized in disease management program" — justify / What does biological control mean? Discuss the mechanism of antagonism

■ Source: Original Course Content

Answer

1. Definition of Biological Control

Biological control is the suppression of a plant pathogen population by another organism (referred to as the antagonist/biocontrol agent). It is a population-leveling process in which the population of the antagonist lowers the survival or density of the pathogen. First coined by **Harry Smith**.

2. Why Biological Control is Prioritized:

- **Eco-friendly:** Chemical pesticides are highly toxic to non-target beneficial organisms, leave residues, and cause massive soil/water pollution. Biocontrol agents are non-toxic and biodegradable.
- **Pathogen Resistance:** Pathogens frequently develop rapid resistance to single-site chemical fungicides. Development of resistance to multi-site biocontrol mechanisms is highly unlikely.
- **Sustainability:** Chemical pesticides require repeated treatments, while established biocontrol populations can self-replicate and persist in the rhizosphere for years.

3. Mechanisms of Antagonism (How BCA Suppresses Pathogen):

- **A. Antibiosis:** Inhibition of one organism by another through the secretion of antibiotics or toxic metabolites.
- *Example:* *Bacillus cereus* produces *Zwittermicin A* against *Phytophthora* root rot in alfalfa; *Bacillus subtilis* produces *Iturin A* against *Botrytis cinerea*.
- **B. Nutrient Competition:** Direct competition between BCA and pathogen for vital resources like Carbon, Nitrogen, Oxygen, or Iron.
- *Example:* *Pseudomonas fluorescens* prevents bacterial blotch by competing with *Pseudomonas tolaasii*.
- **C. Mycoparasitism / Hyperparasitism:** The antagonist directly invades and feeds on the pathogen hyphae.
- *Example:* *Trichoderma harzianum* coils around host hyphae, secretes cell-wall-degrading enzymes (chitinases, glucanases), and penetrates *Fusarium culmorum*.
- **D. Lytic Enzymes:** Release of enzymes (chitinase, β -1,3-glucanase, protease) that directly hydrolyze and rupture pathogen cell walls.
- **E. Siderophore Secretion:** BCA releases high-affinity iron chelators (**siderophores**) that bind free Fe^{3+} , making it unavailable to rhizosphere pathogens.
- **F. Induced Systemic Resistance (ISR):** Colonization of roots by PGPRs (*Pseudomonas*, *Trichoderma*) induces plant systemic resistance depending on ethylene and jasmonic acid pathways.
- **G. Cross Protection:** Inoculation with a mild virus strain protects the host against severe virulent strains of the same virus (e.g., Citrus Tristeza Virus).

Q7: What is host resistance? Delineate the physical method of seed treatment with an example

■ Source: Original Course Content

Answer

Host Resistance: The ability of an organism to exclude or overcome pathogens completely or in some degree. Includes true resistance (gene controlled — vertical and horizontal) and apparent resistance (disease escape or disease tolerance).

Physical method of seed treatment — Hot Water Treatment:

- Pre-soaking the seed
- Pre-heating at 10°C for 1–2 min
- Hot water soak at appropriate temperature
- Drying
- Cooling

Example — Tomato seed protocol:

- Preheat seed for 10 min at 100°F (37°C)
- Transfer to 122°F (50°C) for 25 min
- Cool immediately and dry completely prior to sowing or re-packaging
- May be treated with fungicidal dust when dry
- For mung bean: Hot water at 55°C/20 min, 60°C/10 min, 70°C/5 min, or 80°C/2 min — none reduced germination below 95%.

Q8: [♦ Repeated ×2] Suggest the best cultural technique with its mechanism for the control of late blight of potato when it appears every year on same piece of land

■ Source: Original Course Content

Answer

- Late blight is caused by *Phytophthora infestans* — a polycyclic pathogen.
- Best cultural technique: Crop Rotation

Mechanism:

- *P. infestans* survives in infected tubers (vegetative mycelium in tubers = dormant inoculum)
- Growing potato every year in the same land allows the pathogen to persist and build up inoculum (X■)
- Rotating with a non-host crop (e.g., wheat, rice) for 1–2 seasons starves the pathogen of its host
- This reduces initial inoculum (X■) and delays epidemic onset

Supporting measures:

- Sanitation: Remove and destroy infected crop residues and cull piles after harvest (eliminates overwintering inoculum)
- Selection: Use certified, disease-free seed tubers
- Chemical protection if needed: Mancozeb + Metalaxyl (Ridomil Gold) or Copper Oxychloride (Blitox)
- Why this is best: Since late blight is a polycyclic disease, reducing initial inoculum (X■) combined with removing the substrate from the same field directly breaks the disease cycle and prevents the sigmoid epidemic from initiating.

Q9: [♦ Repeated ×3] Differentiate vertical resistance from horizontal resistance. Which one do you prefer and why?

■ Source: Original Course Content

Answer

Comparison: Vertical vs. Horizontal Resistance

Feature	Vertical Resistance (VR)	Horizontal Resistance (HR)
Definition	Complete resistance to some races of a pathogen but not to others.	Partial resistance equally effective against all races of a pathogen.
Gene Basis	Monogenic or oligogenic (major genes).	Polygenic (minor genes).
Host Specificity	Race-specific / Host specific.	Race non-specific / Host non-specific.
Effect on Pathogen	VR does not allow the pathogen to establish in the host — no increase of disease.	HR allows the pathogen to establish but decreases the rate of disease increase.
Epidemiological Implication	Epidemic cannot develop until breakdown by a new pathogen race.	Epidemic development depends on the level of HR and environmental conditions.
Durability	Less durable — can be rapidly overcome by new virulent races.	More durable.

Preference & Justification:

We strongly prefer Horizontal Resistance (HR) in broad-scale agricultural systems for the following reasons:

1. **Durability:** Since it is governed by multiple minor genes (polygenic), the pathogen cannot overcome it with a single nucleotide mutation. Thus, it lasts much longer in the field.
2. **Stability:** It provides stable, race-non-specific protection. Unlike the "boom and bust" cycle of Vertical Resistance, there is no threat of sudden complete crop failure when new pathogen races evolve.
3. **Epidemic Suppression:** Although it allows the pathogen to establish at minor, non-damaging levels, it directly slows down the rate of epidemic progress (r), keeping disease severity below economic injury levels.

Q10: "Mixed cropping reduces disease incidence" — briefly discuss the reasons

■ Source: Original Course Content

Answer

- Reduces the density of any single host species in a field, lowering the chance of rapid pathogen spread from plant to plant
- Acts similarly to crop rotation in disrupting the disease cycle
- Genetically diverse host population reduces the effectiveness of host-specific pathogens
- Physical barriers between susceptible plants reduce wind and rain-splash dissemination of inoculum

Reduces genetic uniformity — genetically uniform (monoculture) host plants are most susceptible to epidemics; mixed cropping increases genetic diversity, helping to develop epidemic resistance

Q11: List two advantages and limitations of biological control agents in plant disease control

■ Source: Original Course Content

Answer***Advantages:***

- Non-toxic to human; not a water contaminant concern; safer to environment
- Once colonized may last for years; host specific; also enhances plant growth

Limitations:

- Biocontrol agents are not broad spectrum and are less effective than fungicides; slow effect
- Short shelf life; unavailable in larger quantities; only preventive, not curative; efficiency mainly decided by environmental conditions

Q12: Why do resistant varieties sometimes breakdown in the field?

■ Source: Original Course Content

Answer***From the content on Vertical Resistance (VR) and epidemiological implications:***

- Vertical resistance provides complete resistance to some races of a pathogen but not to others
- When a new race of the pathogen evolves (through mutation or genetic recombination), this new race can overcome the VR of the previously resistant variety
- This is called breakdown of resistance
- Epidemics cannot develop in VR varieties until breakdown by a new pathogen race
- VR does not allow the pathogen to establish, but once a virulent race appears, the host becomes completely susceptible and epidemic can develop even under favorable environment

Q13: Discuss the importance of host resistance within Integrated Disease Management (IDM)

■ Source: Original Course Content

Answer

- Host resistance is one of the 6 basic types of control measures in plant disease management
- Prevents infection or reduces the effect of infection by managing the host through genetic improvement
- Vertical resistance reduces the effective amount of incoming inoculum (avirulent races cannot infect)
- Horizontal resistance decreases the rate of disease development (r) for all races
- Resistant varieties are the most economical, environmentally safe control measure
- They reduce reliance on chemical fungicides, reduce cost of production, and provide season-long protection
- Within IDM, resistant varieties form the backbone while cultural, biological and chemical methods supplement

Q14: Analyze how cultural measures contribute to climate-smart disease management

■ Source: Original Course Content

Answer

- Crop rotation: reduces inoculum levels without any chemical input — sustainable
- Selection of planting date: avoids pathogen peak activity periods — adapts to changing climate patterns
- Adjusted seeding rate and canopy density: reduces humid microclimate within canopy — reduces disease favorable conditions
- Balanced fertilizer application: maintains plant health without promoting excessive susceptible growth
- Sanitation: (burning/plowing residues) reduces overwintering inoculum
- Irrigation management: avoiding overhead irrigation to reduce leaf wetness, which is critical since moisture is the predominant epidemic factor
- Trap crops and cover crops: reduce pathogen movement and provide physical barriers
- These methods collectively reduce initial inoculum (X_0) and rate of epidemic (r) without environmental contamination — making them climate-smart.